BUSINESS ANALYSIS WORK SAMPLE

St. John's Council Minutes Assistant

Business requirements, user stories, UAT plan and traceability for an AI service that answers questions about council decisions, with citations to the official minutes

Author	Mostafa Rafiur Wasib

Version	0.9.1, complete draft for stakeholder validation
Date	17 October 2026 (analysis); 18 October 2026 (Project A's measured results recorded)
Method	BABOK Guide v3 requirements classification; BPMN 2.0; MoSCoW; INVEST and Gherkin; WCAG 2.1 AA
Implementation	Project A: github.com/mrw-soumik/stjohns-council-minutes-rag
Repository	github.com/mrw-soumik/business-analysis-council-minutes-search
Short version	Two-page summary (PDF)

Read it in five minutes

1. BRD executive summary (Part 1, section 1)
2. Current-state and future-state process maps (Diagrams 1 and 2)
3. Requirements with acceptance criteria (Part 1, section 9)
4. Stories US-05, US-07 and US-15 with Gherkin criteria (Part 2)
5. Traceability and measured results: requirement to test to automated check (Part 4, sections 3 and 4)

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About this pack

What this is

The **St. John's Council Minutes Assistant** lets a resident, journalist, community volunteer or City staff member ask a question such as "What did council decide about 351-361 Kenmount Road?". The answer is short and cites the exact meeting, date and page, with a link to the official document. When the minutes do not answer the question, it says "not found in the minutes" instead of guessing.

This repository is the **business analysis** for that product, written to the standard a municipal or consulting business systems analyst would produce. It uses BABOK Guide v3 terminology and requirements classification (business, stakeholder, solution and transition requirements), BPMN 2.0 process models, MoSCoW prioritization, INVEST user stories with Gherkin acceptance criteria, and a requirements traceability matrix.

Who it is for

- **Hiring managers and recruiters** for Business Analyst and Business Systems Analyst roles: a realistic, end-to-end example of requirements work on an AI product in a public-sector setting.
- **Delivery teams:** the requirements and acceptance tests that the implementation (Project A) is built and measured against.

Five-minute path through the full pack

1. **Executive summary:** BRD section 1. The problem, the solution, scope and the decisions needed.
2. **Process maps:** the current state with pain points, then the future state. Both are BPMN 2.0 models.
3. **Requirements:** BRD section 9. Shared IDs (BR, FR, NFR) with priority, rationale and testable acceptance criteria.
4. **Three stories:** US-05 (a deferral must not be reported as an approval), US-07 (citations) and US-15 (ingestion).
5. **Traceability and results:** the forward matrix, from business requirement to the automated check that measures it, and the measured results.

What is in the pack

Document	Contents
00-executive-summary.md and its PDF	Two-page summary: problem, scope, stakeholders, current and future state, the ten most important requirements, success measures, acceptance approach and measured results, top risks and links to everything else
01-business-requirements-document.md	The core BRD: document control, executive summary, background, SMART success measures with baseline methods, scope, stakeholder register, power and interest grid, RACI, elicitation plan, proto-personas, current and future state with BPMN maps, gap analysis, requirements with IDs and acceptance criteria, data requirements and conceptual data model, business rules, transition requirements, assumptions, constraints and dependencies, risks (including AI, privacy and copyright risks), open questions, glossary and a blank sign-off table
02-user-stories.md	7 epics and 26 INVEST user stories with Gherkin acceptance criteria, MoSCoW priority, relative estimates and the requirements each satisfies; prioritization rationale and a two-release plan
03-uat-test-plan.md	UAT objectives, approach, entry and exit criteria, roles, environment, test data and question bank, defect severity and priority, schedule template, 30 test cases (all "Not run"), and Project A's measured results for the 14 automated checks

Document	Contents
04-traceability-matrix.md and traceability.csv	Business requirement to functional or non-functional requirement to user stories to UAT test cases to Project A automated checks, with each requirement's measured status; generated from the documents and checked for gaps
diagrams/	Current-state and future-state BPMN 2.0 models (`.bpmn`) with SVG and PNG renders, the context diagram and the conceptual data model
evidence/ and scripts/portal_snapshot.py	The reproducible source of every current-state count quoted in the BRD

How it connects to Project A

The implementation lives in a separate repository, stjohns-council-minutes-rag (Project A), a retrieval-augmented generation (RAG) application. The two projects share requirement IDs: BR-01 to BR-03, FR-01 to FR-07 and NFR-01 to NFR-07.

- This pack **defines** the requirements and their acceptance thresholds. For example, NFR-01 requires the correct source page in the top five results for at least 85% of answerable test questions.
- Project A **implements** them and **measures** the thresholds with its automated evaluation suite.
- The traceability matrix links each requirement to the Project A check that measures it (PA-01 to PA-14). Project A's measured results were recorded on 18 October 2026: 13 of the 14 checks met their thresholds (PA-02 is report-only). The p95 answer time for NFR-03 (12.82 s against a 10 s target) did not, and the next steps are recorded. Citations use the agenda item instead of the page number because of the PDF access constraint (C-03).

Key findings from the current-state analysis

Observed on the City's public pages and eSCRIBE portal on 17 October 2026 (sources and method in BRD section 7.2 and Appendix A):

- Regular Council Meetings are where council makes its formal decisions. Committee of the Whole makes no final decisions, so the first release covers Regular Council Meeting minutes only.
- The portal lists 253 Regular Council Meetings from 8 July 2019 to 6 October 2026. 242 list minutes; 9 of those are HTML-only, with no page numbers to cite, which led to requirement FR-02.3.
- The portal's keyword search returns meetings and documents but no text excerpts, so each result must be opened and read.
- Minutes are adopted at a later meeting, so the newest decisions are not yet searchable. This drives the "not yet published" message (FR-03.3) and visible coverage dates (NFR-07.2).
- Scripted requests for minutes PDFs received a browser verification page. That makes an agreed access method a dependency and an open question for the City (D-05, OQ-05).

What this pack does and does not claim

- Current-state facts are limited to what is publicly observable, each with its source. Everything about City internal processes is labelled as an assumption to validate.
- No interviews, workshops or surveys have taken place. The personas are proto-personas, and BRD section 5.5 sets out the elicitation plan that would validate them.
- No baseline is invented. Success measures state how each baseline will be measured; targets are proposals for the sponsor to confirm.

- Measured results are Project A's own, copied from its README and `eval/results`; this pack does not re-measure them. No human UAT has been performed, so every UAT test case is still "Not run".

Business Requirements Document: St. John's Council Minutes Assistant

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> **Status of this analysis.** This document is based on public information and document analysis carried out on 17 October 2026. No stakeholder interviews or workshops have been held. Every statement about City processes, enquiry volumes or stakeholder needs that could not be observed publicly is marked as an assumption to validate with stakeholders. Section 5.5 sets out the elicitation plan that would confirm or correct them.

Document control

Item	Value
Document	Business Requirements Document (BRD)
Document ID	CMA-BA-01
Product	St. John's Council Minutes Assistant
Version	0.9.1 (complete draft for stakeholder validation)
Status	Draft, not baselined
Author	Mostafa Rafiur Wasib (business analyst)
Date	17 October 2026
Proposed approvers	Business sponsor, product owner, City Clerk's office representative, ATIPP Coordinator (see section 17)
Related documents	02-user-stories.md (CMA-BA-02), 03-uat-test-plan.md (CMA-BA-03), 04-traceability-matrix.md (CMA-BA-04), implementation: stjohns-council-minutes-rag (Project A)
Method	BABOK Guide v3 requirements classification: business, stakeholder, solution (functional and non-functional) and transition requirements

Version history

Version	Date	Author	Change
0.9	2026-10-17	Mostafa Rafiur Wasib	Complete draft based on public information and document analysis. Elicitation sessions not yet held.
0.9.1	2026-10-18	Mostafa Rafiur Wasib	Recorded Project A's measured results in the UAT plan (section 14) and the traceability matrix. FR-05.1 acceptance wording notes the 80-character excerpt limit that PA-13 checks. No requirement changed.

The next version (1.0) is the baseline, issued after the validation activities in section 5.5 and sign-off in section 17.

1. Executive summary

St. John's City Council makes its formal decisions at Regular Council Meetings. The City publishes the minutes of these meetings on its public eSCRIBE meetings portal. On 17 October 2026 the portal listed 253 Regular Council Meetings held since 8 July 2019, of which 242 list published minutes (evidence: portal snapshot).

The portal has a calendar, a list view and a keyword search with filters. Search results list meetings and documents, not answers, and show no text around the matched words. To find what council decided on a topic, a person opens documents one at a time and searches inside each one. When that fails, people can contact the City or make a formal access to information request. The City's access to information page notes that minutes of Council meetings are available without a formal request.

This document defines the requirements for the **St. John's Council Minutes Assistant**. It is a web service that answers plain-language questions about decisions recorded in Regular Council Meeting minutes. Each answer cites the meeting, date and page and links to the official document. When the minutes do not contain the answer, the service says "not found in the minutes" instead of guessing. The official minutes on eSCRIBE remain the record of authority. The assistant is a way to find them faster, not a replacement.

Key points

- **Business requirements:** reduce the time to find what council decided (BR-01); make every answer traceable to the official record (BR-02); reduce routine "what did council decide about X" enquiries handled manually (BR-03, an assumption to validate).
- **Scope:** Regular Council Meeting minutes published on eSCRIBE (portal coverage starts 8 July 2019). Committee of the Whole, agendas, attachments and video are out of scope for the first release.
- **Requirements:** 11 stakeholder requirements, 7 functional requirements (27 detailed requirements), 7 non-functional requirements (25 detailed requirements), 11 transition requirements and 11 business rules. Each has an ID, MoSCoW priority, rationale and acceptance criteria.
- **Quality gates:** the correct source page is in the top 5 results for at least 85% of answerable test questions. At least 90% of citations support the claim made. At least 80% of unanswerable questions are declined. These thresholds are measured automatically by the implementation (Project A) and traced in 04-traceability-matrix.md.
- **Main risks:**
  - wrong or outdated answers
  - over-reliance on AI output
  - misquoting
  - accessibility of dynamic content
  - privacy under the Access to Information and Protection of Privacy Act, 2015 (ATIPPA, 2015)
  - copyright in City documents
  - continued access to the source portal (scripted requests for PDF files received a browser verification page during this analysis)
- **Decisions needed:**
  - validate the personas and the BR-03 baseline with the City Clerk's office and 311
  - agree an access method for minutes documents
  - confirm the disclaimer wording and the retention period for question logs
  - approve a four-week pilot

2. Background and problem statement

2.1 Context

St. John's City Council conducts its business through two types of public meeting, held on alternating Tuesdays at 3 p.m. (City of St. John's, Council Meetings):

- **Regular Council Meetings** are the formal decision-making meetings. The City describes decisions made at Regular Meetings as "the official position of Council".

- **Committee of the Whole** meetings are for discussion, review and direction. "No final decisions are made at Committee of the Whole"; its recommendations go to a later Regular Council Meeting for approval.

Agendas, minutes, decision notes and video are published through the Council Meeting Calendar, an eSCRIBE portal at pub-stjohns.escribemeetings.com. Minutes record each agenda item with its number and title, the motion, the mover and seconder, the vote and the outcome (for example "MOTION CARRIED (8 to 2)"), and a resolution number such as SJMC-R-2026-09-22/309. The minutes of one meeting are adopted at a later meeting. For example, the 22 September 2026 meeting adopted the minutes of 8 September 2026.

Residents, journalists, community groups and City staff regularly need to know what council decided about a street, a development application, a plan or a policy. The record exists and is public. The difficulty is finding the right passage quickly and being sure it is the latest decision.

2.2 Problem statement

The problem of	finding what council decided on a topic in Regular Council Meeting minutes
affects	residents, journalists and researchers, community association volunteers, and City staff who answer public enquiries
the impact of which is	time spent opening and reading documents one at a time; decisions missed when the minutes use different words; later decisions (deferrals, amendments) overlooked; enquiries to the City, and sometimes formal access requests, for records that are already public
a successful solution would	let people ask in plain language and get a short answer that cites the exact meeting, date and page with a link to the official document, or an honest "not found in the minutes", while the eSCRIBE portal remains the official record

2.3 Business analysis core concept model (BACCM)

Concept	For this initiative
Need	Faster, trustworthy access to council decisions that are already public.
Change	Add a question-answering layer over published minutes that always points back to the official document.
Solution	St. John's Council Minutes Assistant (implemented as a retrieval-augmented generation application in Project A).
Stakeholders	Residents, journalists and researchers, community associations, City Clerk's office, 311, ATIPP Coordinator, City IT, Legal, Communications, Mayor and Council (section 5).
Value	Time saved for the public and staff; more confidence in answers because each one is traceable; fewer avoidable enquiries.
Context	Public records under ATIPPA, 2015; City copyright in its documents; an existing vendor portal (eSCRIBE) that the City controls; AI-specific risks of wrong or overconfident answers.

3. Business objectives and success measures

3.1 Business objectives

ID	Business objective	Business requirement
OBJ-01	Make it faster for residents and staff to find what council decided on a topic.	BR-01
OBJ-02	Make every answer traceable to the official record, so that people can check it.	BR-02
OBJ-03	Reduce routine decision enquiries that staff handle manually by enabling self-service (assumption to validate).	BR-03

3.2 SMART success measures

No baseline exists yet. Each baseline below is measured before the pilot using the method stated. Targets are proposals for the sponsor to confirm once the baseline is known; they are not claims of results.

ID	Measure	How the baseline will be measured	Proposed target	Timeframe	Owner
SM-01 (BR-01)	Median time to find a council decision for a standard set of six scripted tasks (for example, "find what council decided about a named development application"). Task completion rate is recorded alongside.	Moderated usability sessions with eight participants (two per persona) using the current eSCRIBE portal. Time on task and completion recorded per task. Tasks are written from published minutes so that a correct answer exists.	Median time at least 50% lower with the assistant, and a completion rate no lower than the baseline.	End of the four-week pilot (TR-03)	Business analyst
SM-02 (BR-02)	Share of answers that are traceable to the official record: (a) answers other than "not found" with at least one complete citation and a working official link; (b) citation support rate (NFR-02).	Not applicable: current tools return documents, not answers. Recorded as "no answer-level traceability".	(a) 100% of answers; (b) at least 90% of citations support the claim.	Every release (automated checks PA-11 and PA-03)	Product owner
SM-03 (BR-02)	Share of participants who agree or strongly agree with "I could check this answer against the official minutes" (five-point scale).	The same statement, asked about the current portal at the end of each baseline session.	At least 80% agree, and higher than the baseline.	End of pilot	Business analyst
SM-04 (BR-03)	Number of routine "what did council decide" enquiries handled manually per month by the City Clerk's office and 311.	Eight-week tagging exercise before the pilot using an agreed enquiry category. Supplemented by a review of the City's public list of completed access requests for requests about minutes that were already published.	25% fewer per month within six months of launch (to confirm after the baseline; depends on adoption).	Six months after launch	Business sponsor
SM-05 (BR-03)	Average staff handling time for a routine decision enquiry.	Start and end time recorded for at least 30 enquiries during the tagging period.	30% lower when staff use the assistant (to confirm after the baseline).	End of pilot and three months after launch	City Clerk's office supervisor

Monitoring indicators (no target): questions per month (aggregate count, no identifiers), "not found" rate, problem reports per 100 answers, and the share of answers where the user opened the official link. These show adoption and coverage gaps but are not success measures on their own.

4. Scope

4.1 In scope (solution scope)

- Regular Council Meeting minutes published on the City's eSCRIBE portal, in PDF or HTML, for meetings from 8 July 2019 (the earliest meeting on the portal) onward.
- Plain-language questions and short answers with citations to meeting, date and page (or agenda item for HTML-only minutes) and links to the official documents.
- "Not found" responses, a date range filter, verbatim source excerpts of 300 characters or fewer, and hybrid keyword and semantic retrieval.

- A scheduled, repeatable ingestion pipeline with automated quality checks, plus a search-only mode used for the public demo and as a fallback.
- Transparency information (disclaimer, coverage dates, model name), privacy controls and an accessible web interface.
- Transition activities: initial load, pilot, help content, staff enablement, support process.

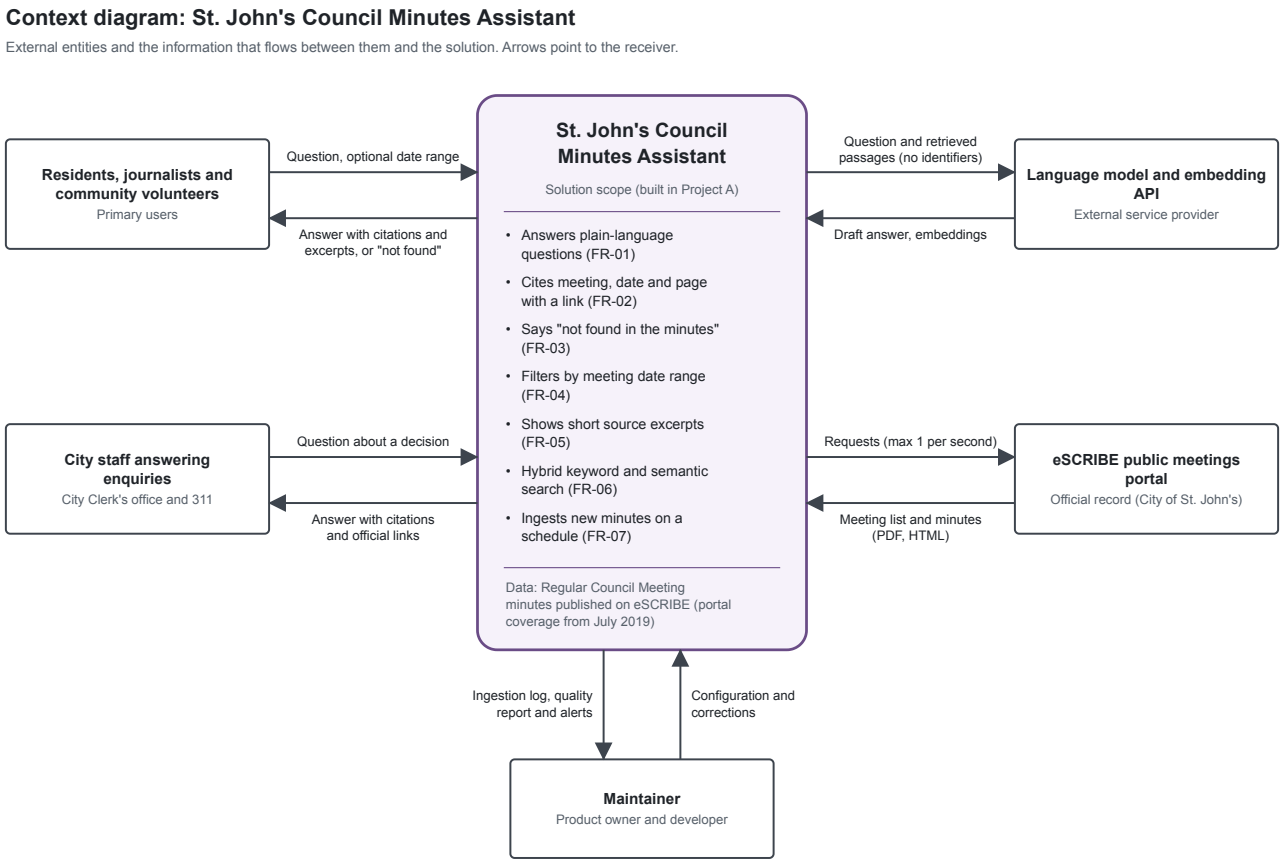
4.2 Out of scope

- Committee of the Whole, standing and advisory committees, working groups and other meeting types (candidates for a later release, OQ-06).
- Agendas, staff reports, decision notes, attachments, presentations and video recordings.
- Minutes not published on eSCRIBE, including any minutes from before July 2019 (OQ-02).
- The City's by-law directory and the full text of by-laws.
- Any change to eSCRIBE or the City's website, and integration with City systems such as a 311 case system.
- User accounts, saved histories and personalized alerts (eSCRIBE already offers meeting notification subscriptions).
- Legal advice, official interpretations, opinions or predictions.
- Languages other than English (minutes are published in English).

4.3 Project scope of this business analysis

This pack covers the business requirements, user stories, UAT plan and traceability for the solution above. The implementation, including its evaluation suite, is Project A. Real elicitation sessions are planned (section 5.5) but have not taken place.

4.4 Context diagram



Full-size version: page 79.

Figure 1. Context diagram. Source: diagrams/context-diagram.svg.

5. Stakeholder analysis

5.1 Stakeholder register

Roles inside the City are inferred from public pages and typical municipal practice. They are assumptions until confirmed (OQ-10).

ID	Stakeholder	Role in this change	Interest	Influence	Key concerns	Engagement
ST-01	Residents	Primary users	High when they need a decision	Low	Ease of use, plain language, being sent to the right document	Involve: usability baseline, pilot, UAT
ST-02	Journalists and researchers	Users who quote the record	High	Medium (public reporting)	Accuracy, exact wording, vote counts, citations	Involve: interviews, UAT
ST-03	Community association volunteers	Users tracking places, plans and applications	High	Low to medium	Completeness over time, dates, addresses and file numbers	Involve: survey, pilot

ID	Stakeholder	Role in this change	Interest	Influence	Key concerns	Engagement
ST-04	City Clerk's office	Assumed owner of council records and minutes; candidate business sponsor	High	High	Accuracy, the official record, workload, how council decisions are portrayed	Manage closely: workshop, sign-off
ST-05	Access St. John's (311) staff	Receive general enquiries	Medium	Medium	Quick, citable answers; clear escalation	Involve: interviews, observation, pilot
ST-06	ATIPP Coordinator	Advises on access and privacy	Medium	High	Personal information, retention, third-party processing, requests for public records	Consult: privacy review, sign-off
ST-07	City IT	Hosting, security and integration if adopted	Medium	High	Security, supportability, vendor terms	Consult
ST-08	Legal (City Solicitor)	Copyright, liability and disclaimers	Low to medium	High	Copyright, disclaimers, reliance on AI output	Consult: disclaimer and terms review
ST-09	Communications	Public messaging	Low to medium	Medium	Consistent messages, expectations	Keep informed
ST-10	Mayor and Council	Their decisions are summarized	Medium	High	Fair and accurate portrayal of decisions	Keep satisfied: briefing before any launch
ST-11	Accessibility advisors (for example the City's Inclusion Advisory Committee)	Accessibility perspective	Medium	Medium	Equal access, assistive technology support	Consult: accessibility testing
ST-12	eSCRIBE vendor	Operates the portal platform	Low	Medium	Load on the portal, terms of use	Monitor
ST-13	Language model provider	External processor of question text	Low	Medium	Terms of use, data handling	Monitor
ST-14	Delivery team (product owner, business analyst, developer)	Builds and runs the solution	High	High	Clear requirements, testable thresholds, cost	Manage closely
ST-15	Office of the Information and Privacy Commissioner (NL)	Independent oversight	Low	High	Compliance with ATIPPA, 2015	Monitor

5.2 Power and interest grid

	Lower interest	Higher interest
Higher influence	**Keep satisfied:** Mayor and Council (ST-10), Legal (ST-08), City IT (ST-07), ATIPP Coordinator (ST-06), OIPC (ST-15)	**Manage closely:** City Clerk's office (ST-04), delivery team (ST-14)
Lower influence	**Monitor:** eSCRIBE vendor (ST-12), model provider (ST-13), Communications (ST-09)	**Keep informed and involve:** residents (ST-01), journalists and researchers (ST-02), community associations (ST-03), 311 staff (ST-05), accessibility advisors (ST-11)

5.3 RACI

R = responsible, A = accountable, C = consulted, I = informed.

Activity	Sponsor (City Clerk's office)	Product owner	Business analyst	Developer	ATIPP Coordinator	City IT	Legal	Communications	311 staff
Approve business requirements	A	C	R	I	C	C	C	I	I
Elicit and validate requirements	C	A	R	I	C	C	I	I	C
Publish and maintain official minutes (unchanged)	A, R	I	I	I	I	I	I	I	I
Build and run ingestion and index	I	A	C	R	I	C	I	I	I
Define and run the evaluation suite	I	A	C	R	I	I	I	I	I
Privacy review (preliminary assessment or PIA)	A	C	C	C	R	C	C	I	I
Accessibility testing	I	A	R	C	I	I	I	I	I
UAT execution and sign-off	A	R	R	C	C	I	I	I	C
Communications and launch	A	C	C	I	I	I	C	R	I
Support and corrections	C	A	C	R	I	C	I	I	C

5.4 Stakeholder requirements summary

The stakeholder requirements (SR-01 to SR-11) that come out of this analysis are in section 9.2. Each one names the stakeholder whose need it expresses.

5.5 Elicitation plan

Technique	Participants	Objectives and sample questions	Output	Status
Document analysis	Author	Review City pages, the eSCRIBE portal, one set of published minutes and the ATIPP pages.	Observed facts (section 7.2), current-state map, Appendix A	Done, 17 October 2026
Interviews (45 minutes each)	City Clerk's office, 311 supervisor, ATIPP Coordinator	How do decision enquiries arrive and how are they answered? How long do they take? How are minutes published, corrected and adopted? What wording must a disclaimer use?	Validated current-state map, BR-03 assumptions, answers to OQ-01, OQ-03, OQ-07, OQ-08, OQ-11	Planned
Observation (job shadowing)	Two staff who answer enquiries	Watch real enquiries end to end (no personal information recorded).	Task timings for SM-05; pain points PP-04 and PP-05 confirmed or corrected	Planned
Moderated usability baseline	Eight participants, two per persona	Six scripted tasks on the current portal; post-task confidence question.	SM-01 and SM-03 baselines; persona validation	Planned
Short online survey	Residents and community associations	How often do you look up council decisions? On which topics? What did you do when you could not find it?	Frequency and topic data; persona refinement	Planned

Technique	Participants	Objectives and sample questions	Output	Status
Requirements workshop (two hours)	Sponsor, City Clerk's office, ATIPP Coordinator, City IT, Legal	Walk through scope, business rules, risks and open questions; agree priorities.	Approved changes; decisions on OQ-05 to OQ-09	Planned
Prototype review	Pilot participants	Review answer format, citations and the "not found" message using the Project A demo.	Feedback on FR-01, FR-02, FR-03, FR-05 wording	Planned (demo available)
Enquiry log sampling	City Clerk's office and 311	Tag decision enquiries for eight weeks.	SM-04 baseline	Planned
Review of completed access requests	Author with the ATIPP Coordinator	Estimate how many requests were about minutes already published (public list).	Evidence for BR-03 and PP-05	Planned

6. Personas

These are proto-personas: assumption-based personas written before research. They are hypotheses to validate in the elicitation activities above, not findings.

P1 Resident

- **Context:** Lives in St. John's and occasionally needs to know what council decided, for example about a development near their home or a change to a service.
- **Goals:** Get a clear answer quickly, in plain language, and know where it came from.
- **Current behaviour (assumed):** Opens the Council Meeting Calendar, guesses a date or runs a keyword search, opens a few documents, then gives up or contacts the City.
- **Pain points:** PP-01, PP-02, PP-03, PP-07.
- **Needs:** SR-01, SR-02, SR-04, SR-09, SR-10.
- **Example question:** "What did council decide about the development at 351-361 Kenmount Road?"
- **Success looks like:** One short answer with the meeting date and a link, or a clear statement that it is not in the minutes.

P2 Journalist or researcher

- **Context:** Reports on or studies council decisions and must quote them accurately.
- **Goals:** Find every relevant decision, the exact wording, the vote and the outcome, and cite the source.
- **Current behaviour (assumed):** Uses keyword search, downloads PDFs and reads them in full to confirm wording and vote counts.
- **Pain points:** PP-01, PP-02, PP-06, PP-08.
- **Needs:** SR-02, SR-03, SR-10.
- **Example question:** "How did the vote go on ratifying the CUPE Local 1289 collective agreement?"
- **Success looks like:** A cited answer with the verbatim excerpt, the recorded outcome and the resolution number, verified in the official PDF before publication.

P3 Community association volunteer

- **Context:** Volunteers for a neighbourhood or community association and follows plans, applications and services in an area over months or years.
- **Goals:** See every decision about a place, plan or application in date order.
- **Current behaviour (assumed):** Keeps personal notes and searches meeting by meeting.

- **Pain points:** PP-03, PP-06, PP-07.
- **Needs:** SR-01, SR-05.
- **Example question:** "What has council decided about the Rennie's River, Central and Mundy Pond neighbourhood plans this year?"
- **Success looks like:** A dated list of decisions with citations, limited to the period asked about.

P4 City staff member answering public enquiries

- **Context:** Works in the City Clerk's office or at 311 and answers questions from the public about past council decisions (assumed).
- **Goals:** Answer accurately and quickly with a reference the caller can check.
- **Current behaviour (assumed):** Repeats the same manual search on eSCRIBE, then replies with the meeting reference or suggests next steps.
- **Pain points:** PP-04, PP-05.
- **Needs:** SR-06, SR-07.
- **Example question:** "Did council approve the request for an extension of development approval at 259 Blackmarsh Road?"
- **Success looks like:** A cited answer found in under a few minutes, copied into a reply with the official link.

7. Current state

7.1 How people find a council decision today

The account below describes only what is publicly observable. Steps inside the City are assumptions and are labelled as such.

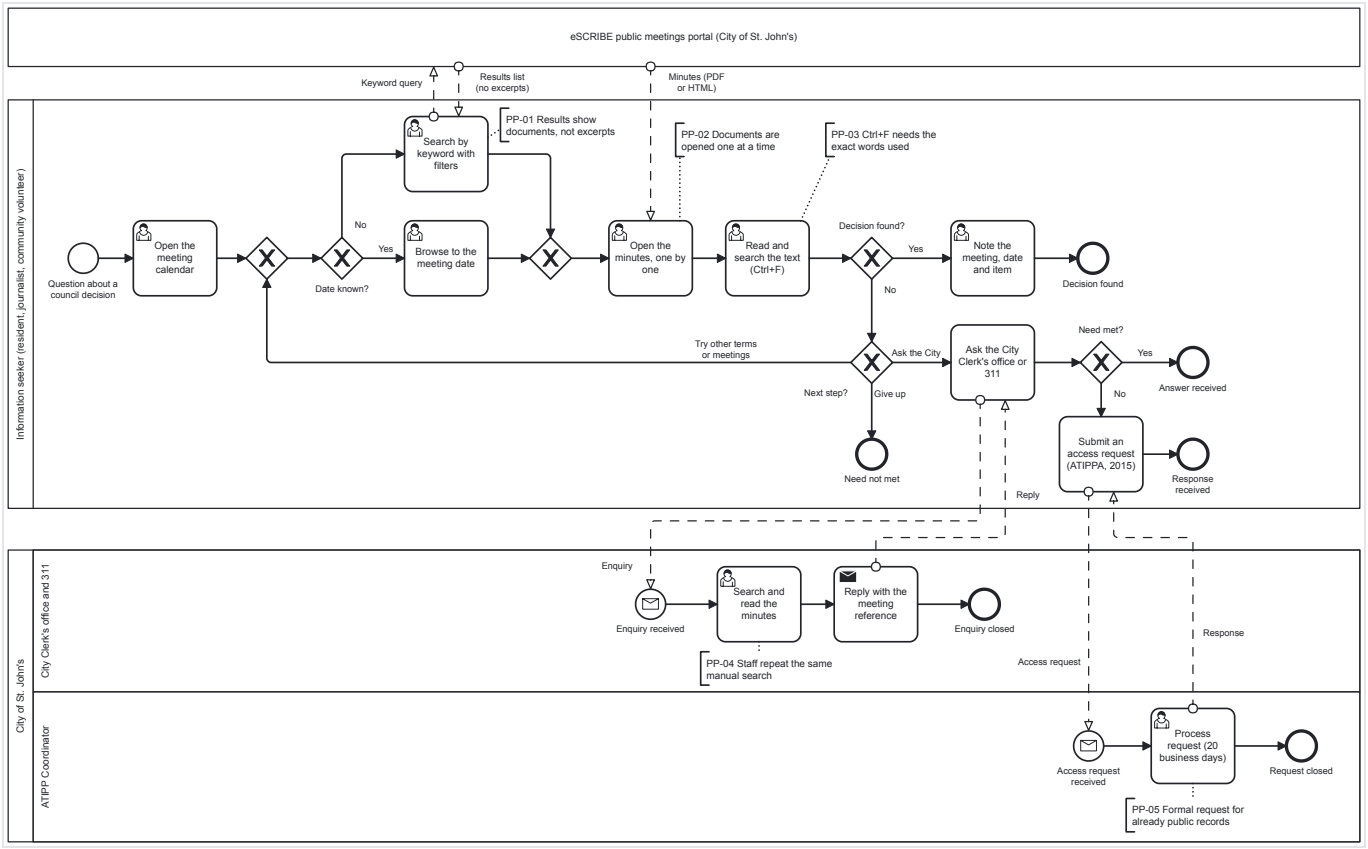
1. A person who wants to know what council decided opens the Council Meeting Calendar. The City's Council Meetings page links to a City page that embeds the eSCRIBE portal.
2. If they know roughly when the meeting took place, they browse the calendar or list. If not, they use the portal's keyword search. It can be filtered by date, meeting type, document type (for example agenda or minutes) and format (PDF or HTML).
3. Search results are grouped by meeting. Each result shows the meeting type, the date, a video link and the matching items or documents. Results do not show the text around the matched words. A test search for "snow clearing", limited to minutes, returned results across many meetings and committees, including items whose titles did not mention the topic. The person cannot tell which result answers the question without opening it.
4. The person opens the minutes, as PDF or HTML, one meeting at a time. They read or use the browser's find function to locate the motion and its outcome.
5. If the decision is found, they note the meeting, date and item. If not, they try other words or meetings, give up, or contact the City by email or phone (City Clerk's office or 311).
6. *Assumed:* City staff repeat the same manual search and reply with a reference or suggest next steps.
7. Some people make a formal access to information request under ATIPPA, 2015. The City asks requesters to send a form to the ATIPP Coordinator, aims to complete requests in 20 business days, and notes that minutes of Council meetings are among the records available without a formal request. *Assumed:* some requests concern minutes that are already published (to check against the City's public list of completed requests).

7.2 Publicly observed facts

All observations were made on 17 October 2026. Counts come from scripts/portal_snapshot.py and the committed evidence/portal-snapshot-2026-10-17.json.

ID	Observation	Source
F-01	Council holds Regular Council Meetings (formal decisions, "the official position of Council") and Committee of the Whole meetings ("No final decisions are made"), on alternating Tuesdays at 3 p.m.	Council Meetings page
F-02	Agendas, minutes, decision notes and video are reached through the Council Meeting Calendar, a City page titled "Council and Committee Minutes and Agendas" that embeds the eSCRIBE portal.	Council Meetings page, City eSCRIBE page
F-03	The portal offers calendar and list views, upcoming and past meetings, and a keyword search with filters for date, meeting type, document type and format (PDF or HTML). It also offers email subscriptions to meeting notices.	eSCRIBE portal
F-04	Search results are grouped by meeting and list matching items and documents with links, but no text excerpts.	Portal search (query "snow clearing", minutes only)
F-05	Each meeting lists its documents, such as Agenda (PDF, HTML), Minutes (PDF, HTML) and Video.	Portal calendar data
F-06	The earliest meeting on the portal is the Regular Council Meeting of 8 July 2019.	Portal snapshot
F-07	253 Regular Council Meetings were held from 8 July 2019 to 6 October 2026. 242 list minutes (233 with a PDF version, 9 in HTML only). 11 list no minutes.	Portal snapshot
F-08	The latest Regular Council Meeting with minutes listed was 22 September 2026. The meeting of 6 October 2026 listed an agenda and video but no minutes. In 2026 to date, 56 meetings were held, 20 of them Regular Council Meetings.	Portal snapshot
F-09	Minutes record each item's number and title, the motion text, mover and seconder, the vote and the outcome, and a resolution number (for example SJMC-R-2026-09-22/309). Minutes are adopted at a later meeting (the 22 September 2026 meeting adopted the minutes of 8 September 2026).	Minutes of 22 September 2026 (HTML)
F-10	Scripted requests for minutes in PDF format received a browser verification page instead of the file. Calendar data and HTML minutes were returned normally.	Observation, method in Appendix A
F-11	The City states that certain records, including minutes of Council meetings, are available without a formal request. Requests go to the ATIPP Coordinator, have no fee to submit, and "should be completed in 20 business days".	Submit an Access Request
F-12	The City publishes a searchable list of completed access requests.	Completed ATIPP Requests
F-13	The City website terms state "Copyright © the City of St. John's (unless otherwise indicated). All rights reserved."	Terms and Conditions of Use
F-14	The portal's robots.txt disallows only one named crawler (PetalBot).	robots.txt

7.3 Current-state process map



Full-size version: page 77.

Figure 2. Current state (as-is), BPMN 2.0. Source model: diagrams/current-state.bpmn (opens in bpmn.io or Camunda Modeler). Pools: information seeker; eSCRIBE portal (black box); City of St. John's with lanes for the City Clerk's office and 311, and the ATIPP Coordinator. City lanes are assumptions.

7.4 Pain points

ID	Pain point	Evidence	Affects
PP-01	Search results show documents, not answers, and no text excerpts, so each result must be opened to be judged.	Observed (F-04)	P1, P2
PP-02	Minutes are opened and read one meeting at a time.	Observed (navigation model, F-05)	P1, P2
PP-03	In-document find needs the exact words used in the minutes. Informal names and paraphrases ("pay deal" for "collective agreement") are missed.	Inferred from how find works; to validate	P1, P3
PP-04	Staff repeat the same manual search when the public asks.	Assumption; to validate	P4
PP-05	A formal access request may be used for records that are already public, which takes the ATIPP Coordinator's time.	Assumption, supported by F-11; to validate with the completed requests list	P1, P4
PP-06	It is hard to follow a topic across meetings and easy to miss a later decision. For example, on 22 September 2026 council deferred one development application for two weeks.	Inferred; example from F-09	P2, P3
PP-07	Coverage gaps are not obvious: recent minutes appear only after adoption at a later meeting, 11 past meetings list no minutes, and the portal holds nothing before July 2019.	Observed (F-06 to F-08)	All
PP-08	Formats differ: 9 meetings have HTML-only minutes, with no page numbers to cite.	Observed (F-07)	P2

7.5 Current-state performance

There is no published measure of how long it takes to find a council decision, or of how many enquiries the City receives about past decisions. These baselines will be measured as described in section 3.2 (SM-01 to SM-05) and are not estimated here.

8. Future state

8.1 Narrative

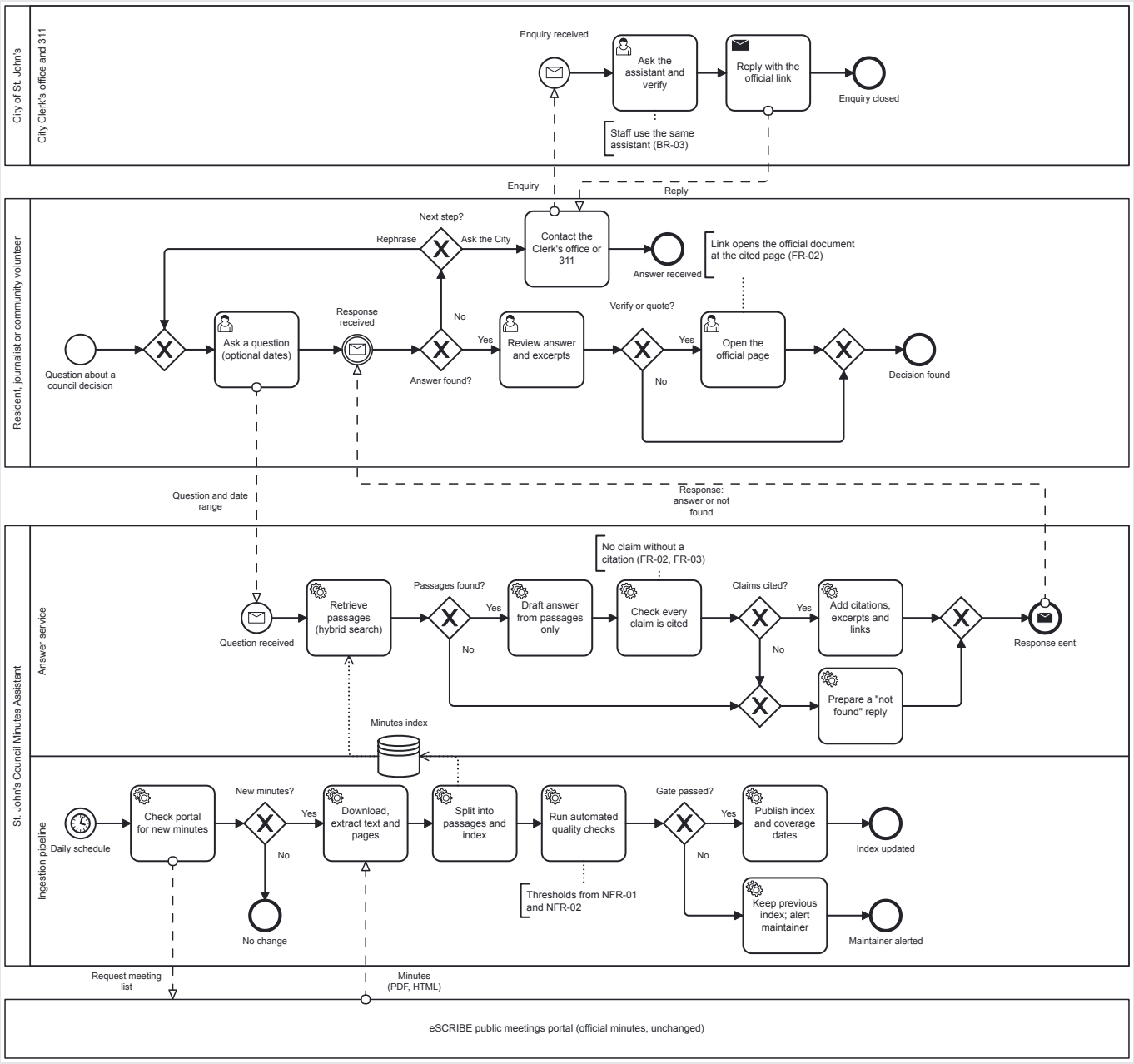
A person types a question in their own words, optionally with a date range. The assistant retrieves the most relevant passages from Regular Council Meeting minutes. It uses keyword search for exact terms such as addresses, file numbers and resolution numbers, and semantic search for paraphrases. It drafts an answer only from those passages and checks that every statement has a citation. It then shows:

- the answer (short, plain language);
- citations in the form "Regular Council Meeting, 22 September 2026, page n", each linked to the official document;
- up to five verbatim excerpts of 300 characters or fewer;
- the coverage period, a disclaimer and the model name.

If the passages do not support an answer, the assistant says so, names the period it searched and suggests next steps.

A scheduled pipeline checks the portal for newly published minutes. It indexes them, runs automated quality checks and publishes the new index only if the quality gate passes. City staff use the same assistant when the public contacts them, check the cited page and reply with the official link. eSCRIBE is unchanged and remains the official record.

8.2 Future-state process map



Full-size version: page 78.

Figure 3. Future state (to-be), BPMN 2.0. Source model: diagrams/future-state.bpmn. Annotations reference requirement IDs.

8.3 What changes for each persona

Persona	Today	Future state	Requirements
P1 Resident	Browses or searches, opens documents one at a time, may give up or contact the City.	Asks in their own words; gets a short cited answer with a link, or a clear "not found" with next steps.	FR-01, FR-02, FR-03, FR-06
P2 Journalist or researcher	Downloads and reads PDFs to confirm wording, votes and outcomes.	Gets the recorded outcome, verbatim excerpts and, where present, the resolution number; still verifies in the official document before quoting.	FR-01.4, FR-02, FR-05

Persona	Today	Future state	Requirements
P3 Community association volunteer	Tracks a place or plan across meetings by hand.	Asks about an address, plan or file number; sees decisions in date order and can limit the period.	FR-01.3, FR-04, FR-06.1
P4 City staff	Repeats the manual search and replies with a reference (assumed).	Uses the assistant, checks the cited page, replies with the official link, and can point people to self-service.	FR-01, FR-02, TR-04

8.4 Gap analysis

Capability	Current state	Future state	Gap closed by
Find a decision without knowing the date	Keyword search lists meetings and documents	Plain-language question returns the decision	FR-01, FR-06
Locate the exact passage	Open each document and use find	Page-level citations and excerpts with the answer	FR-02, FR-05
Handle different wording	Exact words needed	Exact-term and paraphrase retrieval	FR-06
Know when the answer is missing	Not obvious	Explicit "not found" and visible coverage dates	FR-03, NFR-07
Follow a topic over time	Manual reading across meetings	Decisions listed in date order	FR-01.3
Stay current	Portal updated by the City	Index updated automatically after publication	FR-07
Staff answers to enquiries	Manual search (assumed)	Assistant-supported reply with citation	FR-01, FR-02, TR-04

9. Requirements

9.1 Conventions

- **Classification:** BABOK Guide v3. Business requirements (BR) state why. Stakeholder requirements (SR) state what each stakeholder needs. Solution requirements state what the solution must do (functional, FR) and how well (non-functional, NFR). Transition requirements (TR, section 12) state what is needed to move from the current to the future state.
- **IDs:** BR-01 to BR-03, FR-01 to FR-07 and NFR-01 to NFR-07 are shared with Project A and must not be renumbered. Detailed requirements use the parent ID plus a sequence number (for example FR-02.3).
- **Priority (MoSCoW): Must** is needed for the first release; without it the release fails its purpose. **Should** is important but has a workaround or can follow shortly after. **Could** is desirable if capacity allows. **Won't** is agreed out of scope for now (recorded in section 4.2).
- **Acceptance criteria** are written to be testable. Where a UAT case or a Project A automated check verifies them, its ID is given (UAT-nn, PA-nn).

9.2 Business and stakeholder requirements

Business requirements

ID	Business requirement	Priority	Rationale	Measured by
BR-01	Reduce the time it takes residents and staff to find what council decided on a topic.	Must	Current search returns documents, not answers (PP-01 to PP-03).	SM-01
BR-02	Increase trust and transparency: every answer is traceable to the official record.	Must	AI answers are only useful to the public if each one can be checked against the minutes.	SM-02, SM-03

ID	Business requirement	Priority	Rationale	Measured by
BR-03	Reduce routine "what did council decide about X" enquiries handled manually (assumption to validate).	Should	Assumes such enquiries are frequent enough to matter (PP-04, PP-05); confirm with the baseline.	SM-04, SM-05

Stakeholder requirements

ID	Stakeholder	Need	Priority	Traces to	Satisfied by
SR-01	Residents	Ask about a council decision in everyday words, without knowing the meeting date or the wording used in the minutes.	Must	BR-01	FR-01, FR-06
SR-02	All users	Check any answer against the official minutes quickly: meeting, date, page and a working link.	Must	BR-02	FR-02, FR-05
SR-03	Journalists and researchers	Quote the record accurately: see the exact passage, the recorded outcome and where it appears.	Must	BR-02	FR-01, FR-02, FR-05
SR-04	All users	Be told plainly when the minutes do not answer the question, and what to do next.	Must	BR-02	FR-03
SR-05	Community association volunteers	Find every decision about a place, application or plan within a period, in date order.	Should	BR-01	FR-01, FR-04, FR-06
SR-06	City staff answering enquiries	Answer a routine decision enquiry quickly with a citable reference to the official minutes.	Should	BR-03	FR-01, FR-02, NFR-03
SR-07	City Clerk's office	The service never presents itself as the official record and always points to eSCRIBE.	Must	BR-02	FR-02, NFR-07
SR-08	ATIPP Coordinator	No personal information is collected beyond what is necessary, and questions cannot be linked to people.	Must	BR-02	NFR-06
SR-09	People with disabilities	Use the service with a keyboard, screen reader or magnification.	Must	BR-01	NFR-05
SR-10	All users	Know how current the information is and how the answers are produced.	Must	BR-02	NFR-07, FR-07
SR-11	Product owner and maintainer	Keep content current without manual editing, and know when quality drops or costs rise.	Must	BR-01, BR-02, BR-03	FR-07, NFR-01, NFR-02, NFR-04

9.3 Solution requirements: functional

FR-01 Answer natural-language questions about Regular Council Meeting minutes

Priority: Must. **Traces to:** BR-01, BR-03. **Stakeholder requirements:** SR-01, SR-03, SR-05, SR-06.

Rationale: People need the decision, not a list of documents (PP-01 to PP-03).

ID	Requirement	Priority	Acceptance criteria
FR-01.1	Accept a question in plain English of up to 500 characters, with no special syntax.	Must	A question typed as a sentence returns a response. Input over 500 characters is prevented and the limit is stated (UAT-01).
FR-01.2	Return a concise answer (target: 150 words or fewer) written only from the retrieved passages.	Must	Every answer in the UAT question bank is 150 words or fewer and contains no fact missing from its cited passages (UAT-01, UAT-04).
FR-01.3	When several decisions on the topic are found, list them in date order with each meeting date, and say that a later decision may change an earlier one.	Should	For a topic decided at more than one meeting, decisions appear oldest to newest with dates (UAT-10).
FR-01.4	Report the outcome as the minutes record it (for example carried, defeated, deferred, adopted in principle), without upgrading or inferring it.	Must	A deferral is never described as an approval, and "adopt-in-principle" is never described as final adoption (UAT-05).

ID	Requirement	Priority	Acceptance criteria
FR-01.5	Answer only from Regular Council Meeting minutes. If a question is about Committee of the Whole, explain that final decisions are made at Regular Council Meetings and answer from those minutes where possible.	Should	A Committee of the Whole question receives the explanation and, where one exists, the related Regular Council Meeting decision (UAT-09).

FR-02 Every answer cites meeting name, meeting date and page, and links to the official document

Priority: Must. **Traces to:** BR-02. **Stakeholder requirements:** SR-02, SR-03, SR-06, SR-07.

Rationale: Traceability is the main safeguard against wrong AI answers (BR-02, R-01) and keeps the official record in charge (BRL-01).

ID	Requirement	Priority	Acceptance criteria
FR-02.1	Show each citation as meeting name, meeting date and page, for example "Regular Council Meeting, 22 September 2026, page n".	Must	Every citation in the UAT question bank shows all three elements (UAT-02; PA-11).
FR-02.2	Link each citation to the official document on the City's eSCRIBE portal. For PDF minutes, open at the cited page where the browser's PDF viewer supports page links.	Must	The link opens the official document for the cited meeting. The PDF opens at the cited page in supported browsers, or the page number is shown next to the link (UAT-03).
FR-02.3	For minutes published only in HTML, cite the agenda item number and title instead of a page, and link to the HTML minutes.	Should	An answer from one of the nine HTML-only meetings cites "item n" and links to the HTML minutes (UAT-12).
FR-02.4	Support every factual statement in an answer with at least one citation.	Must	Uncited statements count as failures in the NFR-02 evaluation. UAT reviewers find no uncited statements (UAT-04).
FR-02.5	Include the resolution number when the cited passage contains one (for example SJMC-R-2026-09-22/309).	Could	Where the passage contains a resolution number, the answer or citation shows it (UAT-13).

FR-03 When the minutes do not contain the answer, say so instead of guessing

Priority: Must. **Traces to:** BR-02. **Stakeholder requirements:** SR-04.

Rationale: A confident wrong answer is worse than no answer (R-01). Coverage gaps exist (PP-07).

ID	Requirement	Priority	Acceptance criteria
FR-03.1	Show a clear "not found in the minutes" response that names the period searched and offers next steps: rephrase, browse the eSCRIBE portal, or contact the City Clerk's office or 311.	Must	UAT-06. At least 80% of unanswerable evaluation questions are declined (NFR-02; PA-04).
FR-03.2	Decline questions outside council decisions (opinions, predictions, advice, legal interpretation, other organizations) with a one-sentence explanation.	Must	UAT-08.
FR-03.3	When the question or date range is later than the latest minutes available, say that minutes for that period are not yet published and give the latest meeting date covered.	Should	UAT-07.

FR-04 Filter by meeting date range

Priority: Should. **Traces to:** BR-01. **Stakeholder requirements:** SR-05.

Rationale: Volunteers and researchers often ask about a period. A filter reduces noise from older decisions.

ID	Requirement	Priority	Acceptance criteria
FR-04.1	Provide optional start and end dates. When they are blank, search the full coverage period.	Should	UAT-16.
FR-04.2	Validate the range (start on or before end, within coverage) and explain errors in text that assistive technology can read.	Should	UAT-17.

ID	Requirement	Priority	Acceptance criteria
FR-04.3	Apply the filter to the meeting date, not to the publication or retrieval date (BRL-07).	Should	No citation falls outside the selected range (UAT-16; PA-12).

FR-05 Show the retrieved source passages (short excerpts) next to the answer

Priority: Must. **Traces to:** BR-02. **Stakeholder requirements:** SR-02, SR-03.

Rationale: Excerpts let people judge an answer without leaving the page, while respecting City copyright (C-01).

ID	Requirement	Priority	Acceptance criteria
FR-05.1	Show up to five excerpts, each 300 characters or fewer, copied verbatim from the minutes, with an ellipsis where shortened.	Must	UAT-11. Every excerpt is a verbatim substring of its source and within the 300-character limit. PA-13 checks the stricter limit used by the published demo: at most 80 characters.
FR-05.2	Label each excerpt with meeting name, date and page (or item) and link it to the official document.	Must	UAT-11.
FR-05.3	Number the citation markers in the answer and match them to the excerpts.	Should	UAT-11.

FR-06 Hybrid retrieval (keyword + semantic) so exact terms (by-law numbers, street names) and paraphrases both work

Priority: Must. **Traces to:** BR-01. **Stakeholder requirements:** SR-01, SR-05.

Rationale: Exact identifiers fail with semantic search alone, and paraphrases fail with keyword search alone (PP-03).

ID	Requirement	Priority	Acceptance criteria
FR-06.1	Retrieve the passage containing an exact identifier: civic addresses, street names, by-law numbers, resolution numbers (for example SJMC-R-2026-09-22/309) and development file numbers (for example DEV2600106).	Must	UAT-13, UAT-14. The exact-term subset of the evaluation set meets NFR-01 (PA-01).
FR-06.2	Retrieve passages that use different words from the question (for example "pay deal" for "collective agreement").	Must	UAT-15. The paraphrase subset meets NFR-01 (PA-01).
FR-06.3	Combine keyword and semantic results into one ranked list using a method documented by the delivery team.	Must	Design note in Project A. PA-01 reports keyword-only, semantic-only and combined results.

FR-07 Ingest newly published minutes without manual editing (repeatable pipeline)

Priority: Must. **Traces to:** BR-01, BR-02. **Stakeholder requirements:** SR-10, SR-11.

Rationale: Out-of-date content leads to wrong answers (R-02). Minutes are published after adoption at a later meeting (F-09).

ID	Requirement	Priority	Acceptance criteria
FR-07.1	Check the eSCRIBE portal on a schedule (at least weekly; daily preferred) and ingest newly listed Regular Council Meeting minutes.	Must	UAT-18.
FR-07.2	Make ingestion idempotent: re-running it creates no duplicate meetings, documents or passages.	Must	UAT-19; PA-14.
FR-07.3	Write an ingestion log (meetings found, added, unchanged, failed) and update the coverage dates shown to users.	Must	UAT-20.
FR-07.4	Follow portal etiquette: at most one request per second, a descriptive User-Agent, caching, no re-download of unchanged documents, and no attempt to bypass access controls.	Must	Configuration and log review during UAT-18.
FR-07.5	Detect changed documents (by content hash) and re-index them, keeping only the current version searchable.	Should	UAT-19.

9.4 Solution requirements: non-functional

NFR-01 Quality gate (retrieval): correct source page in the top 5 results (hit@5) for at least 85% of answerable evaluation questions

Priority: Must. **Traces to:** BR-01, BR-02. **Stakeholder requirements:** SR-11.

Rationale: An answer can only be as good as the passages retrieved. Hit@5 is measured automatically in Project A.

ID	Requirement	Priority	Acceptance criteria
NFR-01.1	Keep a version-controlled evaluation set of answerable questions (each with its known source page or item) and unanswerable questions. Include exact-term and paraphrased questions, several years and both formats. Target size: at least 50 answerable and 20 unanswerable questions (to agree with the delivery team).	Must	The evaluation set file and its version are named in every report.
NFR-01.2	Run the gate automatically after each ingestion and before each release. A failed gate keeps the previous index live and alerts the maintainer.	Must	UAT-21.
NFR-01.3	Report hit@5 overall and by subset (exact-term, paraphrase, year, PDF or HTML).	Should	PA-01 and PA-02 reports.

NFR-02 Quality gate (answers): at least 90% of answer citations point to a passage that supports the claim; at least 80% of unanswerable questions correctly declined

Priority: Must. **Traces to:** BR-02. **Stakeholder requirements:** SR-02, SR-04.

Rationale: Measures the two failure modes that matter most to trust: unsupported claims and guessing.

ID	Requirement	Priority	Acceptance criteria
NFR-02.1	Judge citation support with a documented method: human review, or an automated judge checked against a human-reviewed sample.	Must	Method documented in Project A. If an automated judge is used, its agreement with human review is reported.
NFR-02.2	Report both rates with the evaluation set version, model name and date.	Must	PA-03 and PA-04 reports.

NFR-03 Performance: p95 end-to-end answer time under 10 seconds; search-only (demo) under 1 second after the page loads

Priority: Should. **Traces to:** BR-01. **Stakeholder requirements:** SR-06.

Rationale: Staff use the assistant while someone is waiting on the phone or by email. Slow answers would push them back to manual search.

ID	Requirement	Priority	Acceptance criteria
NFR-03.1	Measure p95 answer time over an evaluation run (at least 100 requests, or as agreed) and report it with the test conditions.	Should	PA-05.
NFR-03.2	Measure search-only response time in the browser after the page has loaded.	Should	PA-06.
NFR-03.3	Show a progress indicator while an answer is prepared, announced to assistive technology.	Should	UAT-27.

NFR-04 Cost: average model cost per answer under CAD $0.05 (measured from API usage)

Priority: Must. **Traces to:** BR-03. **Stakeholder requirements:** SR-11.

Rationale: The service must cost less to run than the manual handling it helps avoid, and a public demo must not create open-ended spending.

ID	Requirement	Priority	Acceptance criteria
NFR-04.1	Apply a configurable daily spending cap. When it is reached, switch to search-only mode and show a notice.	Must	UAT-28.
NFR-04.2	Calculate cost from the provider's usage records and convert it to CAD at a documented exchange rate and date.	Must	PA-07.

NFR-05 Accessibility: demo UI meets WCAG 2.1 AA (keyboard use, contrast, labels)

Priority: Must. **Traces to:** BR-01, BR-02. **Stakeholder requirements:** SR-09.

Rationale: A public service must be usable by everyone. Dynamically inserted AI answers are a common accessibility failure.

ID	Requirement	Priority	Acceptance criteria
NFR-05.1	Every function works with a keyboard alone, in a logical order, with a visible focus indicator (WCAG 2.1.1, 2.4.3, 2.4.7).	Must	UAT-23.
NFR-05.2	Text contrast is at least 4.5:1, and 3:1 for large text and interface components (WCAG 1.4.3, 1.4.11).	Must	UAT-25; PA-08.
NFR-05.3	Inputs have visible, programmatic labels, and errors are identified in text (WCAG 1.3.1, 3.3.1, 3.3.2, 4.1.2).	Must	UAT-24; PA-08.
NFR-05.4	New answers and status messages are announced to screen readers without moving focus unexpectedly (WCAG 4.1.3).	Should	UAT-24.
NFR-05.5	Content reflows at 320 CSS pixels wide and stays usable at 200% zoom (WCAG 1.4.10, 1.4.4).	Should	UAT-25.
NFR-05.6	Answers use plain language (target: grade 8 reading level or lower, excluding quoted names and titles).	Could	Readability score of the UAT question bank answers.

NFR-06 Privacy: no accounts, no personal data collected; questions are not stored with identifiers (aligned with NL ATIPPA, 2015 principles)

Priority: Must. **Traces to:** BR-02. **Stakeholder requirements:** SR-08.

Rationale: Under section 61 of ATIPPA, 2015, a public body may collect personal information only on limited grounds. The ground relevant to a service like this is information that "relates directly to and is necessary for an operating program or activity". Section 64 requires reasonable steps to protect the personal information a public body holds. The simplest design that meets both is one that collects nothing personal.

ID	Requirement	Priority	Acceptance criteria
NFR-06.1	No user accounts, tracking cookies or third-party analytics.	Must	UAT-26.
NFR-06.2	Application logs store question text, date and quality data only: no IP address, browser fingerprint or other identifier.	Must	UAT-26; PA-09.
NFR-06.3	Delete question logs after a defined retention period (proposed: 90 days, to confirm with the ATIPP Coordinator).	Should	Evidence of the deletion job reviewed in UAT-26.
NFR-06.4	Show a privacy notice next to the question box: do not enter personal information; what is stored and for how long.	Must	UAT-26.
NFR-06.5	Document where question text is processed (provider, region, retention and training terms) and confirm it with the ATIPP Coordinator before any production use.	Must	Privacy note in Project A; confirmation recorded before go-live (TR-09).

NFR-07 Transparency: disclaimer, data coverage dates and model name visible in the UI

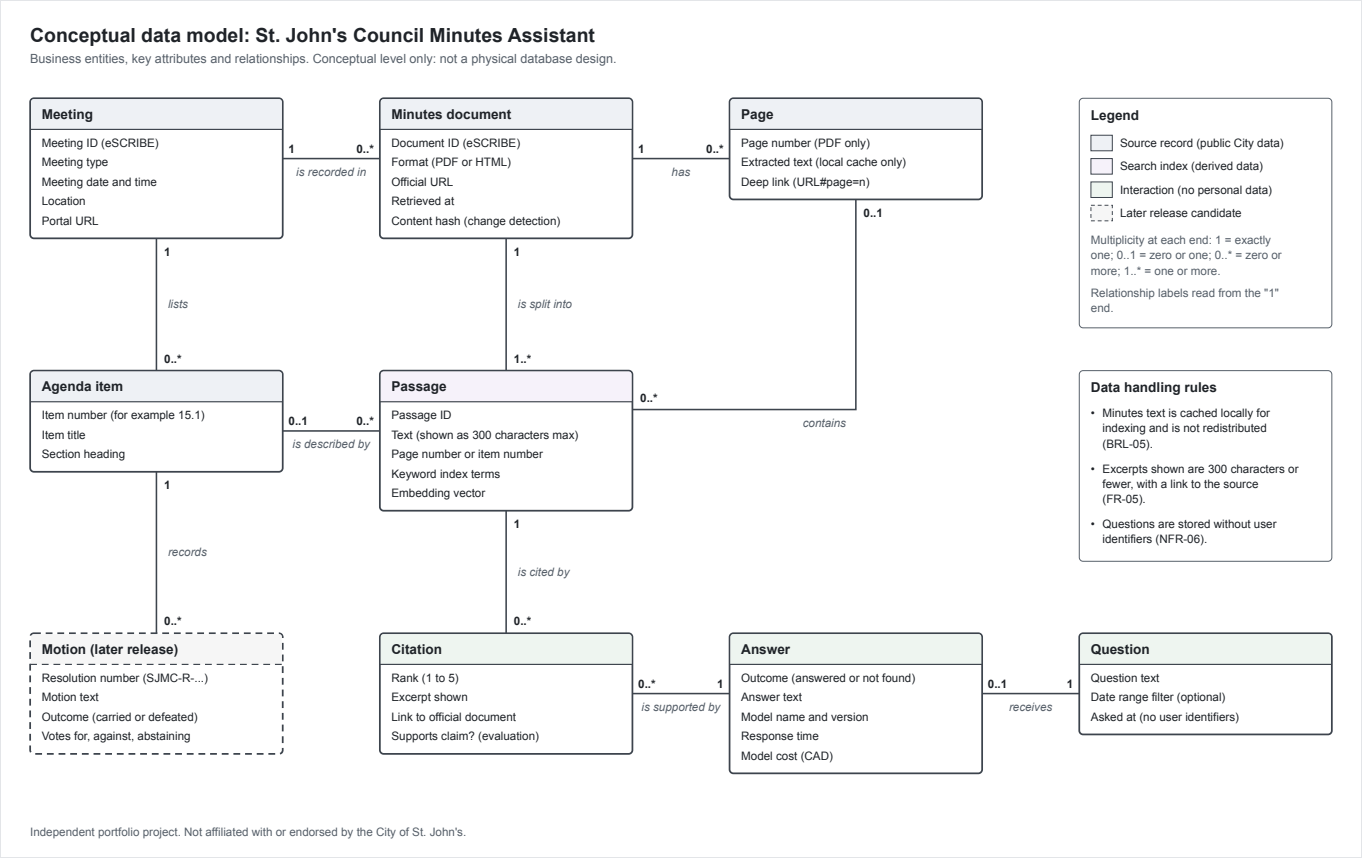
Priority: Must. **Traces to:** BR-02. **Stakeholder requirements:** SR-07, SR-10.

Rationale: Users need to know what the answer is based on, how current it is, and that the official minutes prevail.

ID	Requirement	Priority	Acceptance criteria
NFR-07.1	Show a disclaimer: the service is not the official record, can make mistakes, and the official minutes on eSCRIBE prevail.	Must	UAT-22.
NFR-07.2	Show the coverage period (first and latest meeting included) and the date of the last update.	Must	UAT-20, UAT-22.
NFR-07.3	Show the name and version of the language model used.	Must	UAT-22.
NFR-07.4	Show the independence notice: "Independent portfolio project. Not affiliated with or endorsed by the City of St. John's."	Must	UAT-22.

10. Data requirements and conceptual data model

10.1 Conceptual data model



Full-size version: page 80.

Figure 4. Conceptual data model. Source: diagrams/conceptual-data-model.svg. Conceptual level only; the physical design is a Project A decision.

Relationships

- A Meeting is recorded in zero or more Minutes documents. There are none until minutes are published, and there may be two when both PDF and HTML versions exist.
- A Minutes document has zero or more Pages (PDF only) and is split into one or more Passages.
- A Meeting lists zero or more Agenda items. A Passage belongs to at most one Agenda item and at most one Page.

- An Agenda item records zero or more Motions. Structured motions are a later-release candidate; in the first release, motion text is searched as part of passages.
- A Question receives at most one Answer. The answer's outcome is either "answered" or "not found".
- An Answer is supported by zero or more Citations. It has none only when the outcome is "not found" (BRL-03, BRL-04). Each Citation refers to exactly one Passage.

10.2 Data dictionary

Entity	Attribute	Description	Source	Example or format
Meeting	Meeting ID	Portal identifier of the meeting	eSCRIBE calendar listing	GUID
Meeting	Meeting type	Type as listed on the portal; only "Regular Council Meeting" is in scope	eSCRIBE	Regular Council Meeting
Meeting	Meeting date	Date and start time of the meeting	eSCRIBE	2026-09-22 15:00
Meeting	Portal URL	Link to the meeting page	eSCRIBE	URL
Minutes document	Document ID	Portal document identifier, or the HTML minutes address	eSCRIBE	FileStream DocumentId 51472
Minutes document	Format	PDF or HTML	eSCRIBE	PDF
Minutes document	Official URL	Link used in citations	eSCRIBE	URL
Minutes document	Retrieved at	When the document was last retrieved	Pipeline	ISO 8601 timestamp
Minutes document	Content hash	Fingerprint used to detect changes and prevent duplicates	Pipeline	SHA-256
Page	Page number	1-based page index of the PDF	PDF	4
Page	Extracted text	Text of the page, held in a local cache only and never published	Pipeline	Text
Agenda item	Item number and title	Item as numbered in the minutes	Minutes	15.1, "Ratification of Collective Agreement..."
Motion (later)	Resolution number	Identifier printed in the minutes	Minutes	SJMC-R-2026-09-22/309
Motion (later)	Outcome and votes	Recorded outcome and vote counts	Minutes	Carried, 8 to 2
Passage	Text	Indexed chunk of minutes text; display excerpt limited to 300 characters	Pipeline	Text
Passage	Location	Page number, or item number for HTML-only minutes	Pipeline	page 4 / item 15.1
Passage	Index data	Keyword terms and embedding vector	Pipeline	Vector
Question	Question text	As typed by the user; no identifiers stored with it	User	Text, 500 characters max
Question	Date range	Optional filter on meeting date	User	2025-01-01 to 2025-12-31
Question	Asked on	Date of the question (day precision)	System	2026-10-17
Answer	Outcome	Answered or not found	System	not found
Answer	Answer text	Answer shown to the user	Model and rules	Text, 150 words target
Answer	Model	Model name and version	System	Provider model ID
Answer	Response time and cost	End-to-end seconds and model cost in CAD for this answer	System and provider usage records	Seconds; CAD to four decimal places

Entity	Attribute	Description	Source	Example or format
Citation	Rank	Order of the citation in the answer	System	1 to 5
Citation	Link	Official URL with page anchor where supported	System	URL#page=4
Citation	Supports claim	Evaluation judgement (evaluation runs only)	Evaluation	yes or no

10.3 Data quality rules

ID	Rule
DQ-01	Meeting dates and types come from portal metadata, not from text extracted from documents.
DQ-02	Every passage belongs to exactly one minutes document and one meeting. Orphan passages fail the ingestion run.
DQ-03	Page numbers are the 1-based PDF page index. If printed page labels differ, the index is used and the difference is noted in help content (A-06).
DQ-04	Document ID plus content hash prevents duplicates. A changed hash replaces the earlier version (FR-07.5).
DQ-05	A document that fails text extraction is logged and excluded. It is never partly indexed.
DQ-06	Coverage dates shown to users are computed from the meetings actually indexed, not from the portal listing.

10.4 Retention and classification

Data	Classification	Retention
Minutes text cache	Public City record; copyright of the City	Kept locally for indexing only; never published or committed to a repository (C-01)
Search index and excerpts	Derived from public records	While the service runs; rebuilt from source
Question and answer logs	Operational, no personal identifiers	Proposed 90 days (NFR-06.3, OQ-08)
Evaluation sets and reports	Operational	Kept for audit and comparison between releases

11. Business rules

ID	Business rule	Applied by
BRL-01	The official minutes on the City's eSCRIBE portal are the authoritative record. Where an answer and the minutes differ, the minutes prevail.	FR-02, NFR-07
BRL-02	Answers come only from Regular Council Meeting minutes. Committee of the Whole makes no final decisions; its recommendations go to a later Regular Council Meeting.	FR-01.5
BRL-03	Every factual statement in an answer must be supported by a cited passage.	FR-02.4, NFR-02
BRL-04	If no retrieved passage supports an answer, the response is "not found in the minutes". The assistant does not guess.	FR-03
BRL-05	City documents are not redistributed. Excerpts are verbatim, at most 300 characters each, and always link to the official document.	FR-05.1, C-01
BRL-06	Only minutes listed on eSCRIBE count as evidence of a decision. Agendas, staff reports and video are not used as evidence in this release.	Scope (section 4)
BRL-07	Date filters and coverage dates use the meeting date.	FR-04.3, NFR-07.2

ID	Business rule	Applied by
BRL-08	When several decisions on a topic exist, they are presented in date order, and the answer notes that a later decision, or minutes not yet published, may change the outcome.	FR-01.3, FR-03.3
BRL-09	Outcomes are reported as recorded (for example carried, defeated, deferred, adopted in principle). Vote counts and resolution numbers are quoted only as they appear.	FR-01.4, FR-02.5
BRL-10	No personal information is requested. Questions are stored without identifiers and deleted after the retention period.	NFR-06
BRL-11	Each answer is shown with the coverage period, the disclaimer and the model name.	NFR-07

12. Transition requirements

ID	Transition requirement	Priority	Acceptance criteria	Owner
TR-01	Initial load: ingest all Regular Council Meeting minutes listed on eSCRIBE up to the cut-over date, and reconcile counts with the portal listing (method in Appendix A).	Must	Meetings with minutes listed equal meetings ingested, or each difference is explained in a reconciliation report.	Maintainer
TR-02	Launch baseline: run the full evaluation suite on the initial index and publish the results for NFR-01 to NFR-04.	Must	Report published with the evaluation set version, model and date.	Product owner
TR-03	Pilot: four-week soft launch with representatives of each persona, a short feedback survey, and the SM-01 and SM-03 measurements.	Should	Pilot report with findings and go-forward decisions.	Business analyst
TR-04	Staff enablement: a 30-minute walkthrough and a one-page quick reference for staff who answer enquiries (verifying citations, limits, what to do after "not found").	Should	Material reviewed by the City Clerk's office; sessions held for all pilot staff.	Business analyst
TR-05	Public help content: "How to use", "What answers are based on", "Limitations" and "Privacy" pages in plain language.	Must	Pages published and checked for plain language and accessibility.	Product owner
TR-06	Communications: launch message and FAQ with consistent key messages: a way to find minutes, not the official record; check the link; how to report a problem.	Should	Approved by Communications and the City Clerk's office.	Communications
TR-07	Support: a "report a problem" channel that needs no personal information, a named triage owner, and targets (proposed: acknowledge within 2 business days; correct or suppress a wrong answer within 5 business days).	Must	Process documented; a test report handled end to end (UAT-30).	Product owner
TR-08	Retire workarounds: list the informal aids used today (assumed: bookmarks, personal lists of frequently requested decisions, saved searches) and retire those the assistant replaces after the pilot. eSCRIBE remains the system of record and is not retired.	Could	Inventory and decision log agreed with the City Clerk's office.	City Clerk's office
TR-09	Go/no-go and rollback: go-live requires the UAT exit criteria (UAT plan section 5) and the privacy confirmation (NFR-06.5). Rollback restores the previous index or switches to search-only mode within one hour.	Must	Rollback rehearsed once before go-live.	Product owner
TR-10	Benefits review: repeat the SM-01 to SM-05 measurements at the end of the pilot and three and six months after launch.	Should	Benefits report to the sponsor.	Business analyst
TR-11	Handover: a runbook for ingestion, evaluation and incident response, kept with the code.	Should	A second person can run the pipeline using only the runbook.	Maintainer

13. Assumptions, constraints and dependencies

Assumptions (each to be confirmed or replaced through elicitation)

ID	Assumption	If wrong
A-01	The four personas represent the main user groups and their needs.	Personas, priorities and stories change.
A-02	The City Clerk's office and 311 receive routine enquiries about past council decisions; the volume is unknown.	BR-03 weakens; value rests on BR-01 and BR-02.
A-03	Minutes listed on eSCRIBE are the adopted minutes (supported by F-09).	Answers may need a "draft minutes" label.
A-04	The PDF and HTML minutes of a meeting have the same content.	Citations need a rule for which version prevails.
A-05	Users reach the service with current browsers on desktop and mobile.	Additional compatibility testing.
A-06	PDF page index and printed page labels mostly match.	Help content explains page numbering (DQ-03).
A-07	A hosted model API is acceptable for the portfolio demo; production use would need a privacy review.	Self-hosted model options considered.
A-08	English-only service is acceptable.	Add language support to scope.

Constraints

ID	Constraint
C-01	Copyright: City website terms state "All rights reserved". City documents are not redistributed. The demo shows excerpts of 300 characters or fewer with links to the official document.
C-02	Portal etiquette: at most one request per second, a descriptive User-Agent, caching and no re-fetching of unchanged documents. The portal's robots.txt disallows only PetalBot.
C-03	Scripted requests for PDF minutes received a browser verification page (F-10). Ingestion must work within the portal's access controls and must not bypass them.
C-04	No paid services without approval for the portfolio build. The cost target is NFR-04.
C-05	No user accounts or personal data (NFR-06).
C-06	No City logos, colours or branding. The independence notice is shown everywhere City material is used.

Dependencies

ID	Dependency	Impact if not met
D-01	eSCRIBE portal availability and stable listing structure	Ingestion fails; coverage becomes stale (R-08)
D-02	Model and embedding API availability, pricing and versions	Answers unavailable or more costly (R-09, R-10)
D-03	Project A evaluation results for each release (first results recorded 18 October 2026)	Quality gates unverified; go-live blocked
D-04	Stakeholder availability for elicitation, pilot and UAT	Assumptions remain unvalidated
D-05	An agreed access method for minutes PDFs (OQ-05)	Page-level citations limited to HTML minutes or manual downloads

14. Risks and mitigations

Likelihood and impact are rated Low, Medium or High and combined on a 3 by 3 matrix. The rating is High when one factor is High and the other at least Medium. It is Medium when both are Medium, or when one is High and the other Low. Otherwise it is Low.

ID	Risk	Category	Likelihood	Impact	Rating	Mitigation	Owner	Related
R-01	Wrong answer: the model states something the minutes do not say.	AI quality	Medium	High	High	Answers only from retrieved passages; claim-level citation check; decline when unsupported; NFR-02 gate; disclaimer; report-a-problem channel.	Product owner	FR-02, FR-03, NFR-02
R-02	Outdated answer: a later decision supersedes the one cited, or recent minutes are not yet published (for example, a matter deferred on 22 September 2026 whose next meeting had no minutes listed on 17 October 2026).	AI quality, data	High	Medium	High	Show meeting dates and the coverage end date; list decisions in date order; daily ingestion; "not yet published" message.	Product owner	FR-01.3, FR-03.3, FR-07, BRL-08
R-03	Over-reliance: users treat the answer as the official record or as legal advice.	AI use	Medium	High	High	Disclaimer; citations and excerpts on every answer; staff training; wording such as "According to the minutes of...".	Business sponsor	NFR-07, TR-04, TR-05
R-04	Misquoting: an excerpt is shortened misleadingly, or a vote count or outcome is restated wrongly.	AI quality	Medium	Medium	Medium	Verbatim excerpts with ellipses; outcomes as recorded; no paraphrase inside quotation marks; UAT-05 and UAT-11.	Product owner	FR-01.4, FR-05.1, BRL-09
R-05	Accessibility barrier: dynamic answers not announced to screen readers; some source PDFs may not be accessible.	Accessibility	Medium	High	High	NFR-05 with manual testing by assistive technology users; live regions; link to HTML minutes where available.	Product owner	NFR-05
R-06	Privacy: users type personal information into questions; logs or the model provider keep it.	Privacy	Medium	High	High	No identifiers; privacy notice; short retention; provider terms reviewed; preliminary privacy assessment with the ATIPP Coordinator. Section 72 of ATIPPA, 2015 (privacy impact assessment) is addressed to provincial ministers and departments, so a voluntary assessment is proposed as good practice.	ATIPP Coordinator	NFR-06, OQ-08, OQ-09
R-07	Copyright: reproducing City documents beyond what is permitted.	Legal	Low	High	Medium	No redistribution; excerpts of 300 characters or fewer; links to source; seek City permission before any production use (OQ-15).	Business sponsor	C-01, BRL-05
R-08	Source access: the portal blocks or changes programmatic access (a browser verification step was observed for PDF requests).	Operational	High	High	High	Use HTML minutes where possible; request an agreed access method or open data feed; monitor ingestion failures; never bypass controls.	Maintainer	C-03, D-01, OQ-05

ID	Risk	Category	Likelihood	Impact	Rating	Mitigation	Owner	Related
R-09	Cost overrun from heavy use or abuse.	Financial	Medium	Medium	Medium	Daily spending cap with search-only fallback; rate limiting; monthly cost report.	Product owner	NFR-04
R-10	Model change or retirement alters answer behaviour.	Operational	Medium	Medium	Medium	Pin model versions; re-run the evaluation suite before any change; show the model name.	Maintainer	NFR-02, NFR-07.3
R-11	Reputational: a misrepresented decision is attributed to Council.	Reputational	Medium	High	High	Independence notice; correction targets in TR-07; brief the City Clerk's office before any public promotion.	Business sponsor	TR-06, TR-07
R-12	Prompt injection or misuse: users try to make the assistant ignore its rules.	Security	Medium	Medium	Medium	Scope restriction; refusal rules; output must contain valid citations; no tools or actions available to the model; monitoring.	Developer	FR-03.2, FR-02.4
R-13	Uneven quality across years, topics or formats (for example HTML-only minutes, tables).	AI quality	Medium	Medium	Medium	Evaluation set stratified by year and format; subset reporting (NFR-01.3).	Product owner	NFR-01
R-14	Low adoption: staff or residents keep using the old route.	Adoption	Medium	Medium	Medium	Involve staff in the pilot; measure SM-01 to SM-05; communicate limits honestly.	Business sponsor	TR-03, TR-06
R-15	Questions about private individuals named in minutes (for example applicants or delegations).	Privacy, ethics	Medium	Medium	Medium	Keep answers to decisions; do not compile information about private individuals across meetings; confirm the approach with the ATIPP Coordinator.	ATIPP Coordinator	NFR-06, FR-03.2

15. Open questions for stakeholders

ID	Question	Why it matters	Ask
OQ-01	How many enquiries about past council decisions do the City Clerk's office and 311 receive per month, and are they categorized?	Baseline for BR-03 (SM-04)	City Clerk's office, 311
OQ-02	Where are Regular Council Meeting minutes from before July 2019 published, and should they be in scope?	Coverage (portal starts 8 July 2019)	City Clerk's office, City Archives
OQ-03	Why do 11 past Regular Council Meetings list no minutes on eSCRIBE (for example 11 March 2025 and 6 October 2026)? Pending adoption, published elsewhere, or not produced?	Coverage messages and FR-03.3	City Clerk's office
OQ-04	For the nine meetings with HTML-only minutes, is an agenda-item citation acceptable in place of a page number?	FR-02.3	City Clerk's office
OQ-05	Can the City provide an agreed method for programmatic access to minutes PDFs (open data feed, API or permission), given the browser verification step?	FR-07, R-08	City IT, City Clerk's office
OQ-06	Should Committee of the Whole minutes be added in a later release?	Scope roadmap	Business sponsor
OQ-07	What disclaimer wording do the City Clerk's office and Legal require?	NFR-07.1	City Clerk's office, Legal
OQ-08	What retention period for question logs is acceptable, and is a preliminary privacy assessment expected?	NFR-06.3, R-06	ATIPP Coordinator

ID	Question	Why it matters	Ask
OQ-09	Is processing question text with an external AI provider acceptable? Are there data residency expectations?	NFR-06.5	ATIPP Coordinator, City IT
OQ-10	Who would own the service if the City adopted it?	Sponsor and RACI	City leadership
OQ-11	How are corrections to adopted minutes published?	FR-07.5, R-02	City Clerk's office
OQ-12	Does the City apply an accessibility standard beyond WCAG 2.1 AA (for example WCAG 2.2)?	NFR-05	Accessibility and Inclusion
OQ-13	Should answers include vote counts and resolution numbers when present?	FR-02.5, BRL-09	City Clerk's office
OQ-14	Are decision notes useful supplementary sources for a later release?	Scope roadmap	City Clerk's office
OQ-15	Would the City permit a public demo that shows excerpts of up to 300 characters, or should it show links only?	C-01, FR-05	Legal

16. Glossary

Term	Meaning
ATIPPA, 2015	Access to Information and Protection of Privacy Act, 2015 (SNL 2015, c. A-1.2), Newfoundland and Labrador's access and privacy law for public bodies, including municipalities.
ATIPP Coordinator	City role that receives and processes access to information requests.
BABOK	A Guide to the Business Analysis Body of Knowledge, version 3 (IIBA).
BACCM	Business Analysis Core Concept Model: need, change, solution, stakeholder, value, context.
BPMN	Business Process Model and Notation 2.0, the notation used for the process maps.
Citation	Reference to the meeting name, meeting date and page (or item) that supports an answer, with a link to the official document.
Committee of the Whole	Meeting of all members of Council for discussion and direction; it makes no final decisions.
Coverage period	First and latest meeting dates included in the index.
eSCRIBE	Meeting management platform the City uses to publish agendas, minutes and video.
Excerpt	Verbatim text of 300 characters or fewer, shown next to an answer.
Hallucination	A model statement that is not supported by its sources.
Hit@5	Share of answerable test questions for which the correct source page is among the top five retrieved passages.
Hybrid retrieval	Combining keyword (exact term) and semantic (meaning-based) search.
MoSCoW	Prioritization scheme: Must, Should, Could, Won't.
p95	95th percentile: 95% of measured requests are at or below this time.
Passage	Chunk of minutes text that is indexed and retrieved.
RACI	Responsible, accountable, consulted, informed.
RAG	Retrieval-augmented generation: retrieving source passages and generating an answer only from them.
Regular Council Meeting	Formal decision-making meeting of City Council.
Resolution number	Identifier printed with a motion in the minutes, for example SJMC-R-2026-09-22/309.
Search-only mode	Mode that returns ranked passages with citations without generating an answer.
UAT	User acceptance testing.
WCAG 2.1 AA	Web Content Accessibility Guidelines 2.1, conformance level AA.

17. Sign-off

By signing, the approver confirms that this document correctly states the requirements for the first release, subject to the open questions in section 15.

Role	Name	Decision (approved, approved with changes, not approved)	Signature	Date
Business sponsor				
Product owner				
City Clerk's office representative				
ATIPP Coordinator				
Business analyst (author)				

Appendix A: Current-state observation log

Method. On 17 October 2026 the author reviewed the City's public web pages and the eSCRIBE portal in a browser and with scripted requests. The scripted requests sent at most one request per second, with the descriptive User-Agent "stjohns-council-minutes-rag/0.1 (portfolio research; contact mostafa.wasib73@gmail.com)".

Step	What was checked	Result
1	City Council Meetings page	Meeting types, schedule, link to the Council Meeting Calendar (F-01, F-02)
2	eSCRIBE portal home page and robots.txt	Views, search filters, subscriptions; robots.txt disallows only PetalBot (F-03, F-14)
3	Portal keyword search ("snow clearing", minutes only)	Results grouped by meeting, links only, no excerpts (F-04)
4	Portal calendar listing, 2019 to 2026	Meeting and document counts (F-05 to F-08); reproducible with `python3 scripts/portal_snapshot.py --as-of 2026-10-17`
5	HTML minutes of the Regular Council Meeting of 22 September 2026	Structure of items, motions, votes, resolution numbers and adoption of earlier minutes (F-09)
6	Scripted request for the PDF minutes of the same meeting (two attempts, a few seconds apart)	Browser verification page returned instead of the file (F-10). No attempt was made to bypass it.
7	City access to information pages	Process, 20 business day target, minutes available without a formal request, completed requests list (F-11, F-12)
8	City terms and conditions	Copyright statement (F-13)
9	ATIPPA, 2015 on the House of Assembly site	Sections 61, 64 and 72 (used in NFR-06 and R-06)

What was not done. No City documents are stored in this repository. Only one set of minutes was viewed, to understand its structure. No interviews or surveys have been carried out. Counts reflect the portal listing on 17 October 2026 and will change as new minutes are published.

Appendix B: References

- City of St. John's, Council Meetings: https://www.stjohns.ca/your-government/council-meetings/
- City of St. John's, Council and Committee Minutes and Agendas: https://apps.stjohns.ca/escribe/Escribe.aspx
- City of St. John's eSCRIBE public meetings portal: https://pub-stjohns.escribemeetings.com
- City of St. John's, Access to Information and Protection of Privacy: https://www.stjohns.ca/your-government/access-to-information-and-protection-of-privacy/

- City of St. John's, Submit an Access Request: https://www.stjohns.ca/your-government/access-to-information-and-protection-of-privacy/submit-an-access-request/
- City of St. John's, Completed ATIPP Requests: https://apps.stjohns.ca/cityhall/CompletedATIPPRequests.aspx
- City of St. John's, Terms and Conditions of Use: https://www.stjohns.ca/terms-and-conditions-of-use/
- Access to Information and Protection of Privacy Act, 2015, SNL 2015, c. A-1.2: https://www.assembly.nl.ca/legislation/sr/statutes/a01-2.htm
- W3C, Web Content Accessibility Guidelines (WCAG) 2.1: https://www.w3.org/TR/WCAG21/
- IIBA, A Guide to the Business Analysis Body of Knowledge (BABOK Guide) v3
- OMG, Business Process Model and Notation (BPMN) 2.0: https://www.omg.org/spec/BPMN/2.0/

User Stories and Release Plan: St. John's Council Minutes Assistant

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Item	Value
Document ID	CMA-BA-02
Version	0.9 (draft for validation with the delivery team)
Author	Mostafa Rafiur Wasib
Date	17 October 2026
Derived from	01-business-requirements-document.md (requirement IDs FR, NFR, TR)
Verified by	03-uat-test-plan.md and Project A automated checks; see 04-traceability-matrix.md

1. Purpose

This backlog turns the solution requirements in the BRD into user stories that a delivery team can plan, build and test. Each story names who benefits and why, carries Gherkin acceptance criteria, and lists the requirement IDs it satisfies. Some examples in the acceptance criteria come from the published minutes of the Regular Council Meeting of 22 September 2026, so testers can check them against the real record. Page numbers are shown as `<n>` until confirmed against the official PDF during test preparation.

2. Conventions

- **Story format:** As a *persona*, I want *capability*, so that *benefit*. Personas P1 to P4 are defined in BRD section 6. The product owner and maintainer appear in enabler stories that protect quality and operations.
- **Acceptance criteria:** Gherkin (Given, When, Then). Each story has at least one scenario and is covered by at least one UAT case.
- **Priority:** MoSCoW, consistent with the BRD. A story takes the highest priority of the requirements it satisfies, unless noted.
- **Estimates:** relative story points on a modified Fibonacci scale (1, 2, 3, 5, 8, 13). The reference story is US-08 (2 points). These are the author's initial sizes for planning and are to be re-estimated by the delivery team.
- **INVEST:** independent, negotiable, valuable, estimable, small and testable (section 6).

Definition of Ready

- Persona, benefit, acceptance criteria, priority, estimate and requirement IDs are present.
- Test data is identified (UAT question bank items or evaluation set items).
- Dependencies are known, and open questions that affect the story are resolved or explicitly deferred.

Definition of Done

- All acceptance criteria pass, with automated tests added in Project A where the behaviour can be automated.
- The evaluation suite shows no regression against the NFR-01 and NFR-02 thresholds.
- UI changes pass an automated accessibility scan and a manual keyboard check.
- No personal data is introduced (privacy checklist from NFR-06).

- Help content and the runbook are updated where behaviour changed.
- The product owner has accepted the story.

3. Epics

Epic	Name	Goal	Business requirements	Solution requirements
EP-01	Ask and answer	People get a short, accurate answer or an honest "not found".	BR-01, BR-02	FR-01, FR-03
EP-02	Trust and verification	Every answer can be checked against the official record.	BR-02	FR-02, FR-05, NFR-07
EP-03	Find exact terms and filter	Exact identifiers, paraphrases and date ranges all work.	BR-01	FR-04, FR-06
EP-04	Keep the record current	New minutes are added safely without manual editing.	BR-01, BR-02	FR-07
EP-05	Quality gates	Releases meet agreed retrieval and answer quality thresholds.	BR-01, BR-02	NFR-01, NFR-02
EP-06	Accessible, private and affordable	Everyone can use it; nobody is tracked; costs stay capped.	BR-01, BR-02, BR-03	NFR-03, NFR-04, NFR-05, NFR-06
EP-07	Staff support and feedback	Staff can reuse citations; anyone can report a problem.	BR-02, BR-03	FR-02, TR-07

4. Story summary

ID	Story	Persona	Epic	Priority	Points	Release	Satisfies
US-01	Ask a question in plain language	P1 Resident	EP-01	Must	5	R1	FR-01.1, FR-01.2
US-02	Be told when the minutes do not contain the answer	P1 Resident	EP-01	Must	3	R1	FR-03.1, FR-03.3
US-03	Decline questions outside council decisions	P1 Resident	EP-01	Must	2	R1	FR-03.2
US-04	See all decisions on a topic in date order	P3 Community volunteer	EP-01	Should	5	R2	FR-01.3
US-05	Get the outcome exactly as recorded	P2 Journalist or researcher	EP-01	Must	3	R1	FR-01.4
US-06	Understand where final decisions are made	P1 Resident	EP-01	Should	2	R2	FR-01.5
US-07	See citations with meeting name, date and page	P2 Journalist or researcher	EP-02	Must	3	R1	FR-02.1, FR-02.4
US-08	Open the official document at the cited page	P1 Resident	EP-02	Must	2	R1	FR-02.2
US-09	Cite HTML-only minutes by agenda item	P2 Journalist or researcher	EP-02	Should	3	R2	FR-02.3
US-10	Read the source excerpts next to the answer	P2 Journalist or researcher	EP-02	Must	3	R1	FR-05.1, FR-05.2, FR-05.3
US-11	See coverage dates, disclaimer and model name	P1 Resident	EP-02	Must	2	R1	NFR-07.1, NFR-07.2, NFR-07.3, NFR-07.4
US-12	Find by exact identifier	P3 Community volunteer	EP-03	Must	5	R1	FR-06.1, FR-02.5

ID	Story	Persona	Epic	Priority	Points	Release	Satisfies
US-13	Find with my own words	P1 Resident	EP-03	Must	5	R1	FR-06.2, FR-06.3
US-14	Limit the search to a date range	P2 Journalist or researcher	EP-03	Should	3	R2	FR-04.1, FR-04.2, FR-04.3
US-15	Ingest newly published minutes on a schedule	Maintainer	EP-04	Must	8	R1	FR-07.1, FR-07.4
US-16	Re-run ingestion without duplicates	Maintainer	EP-04	Must	3	R1	FR-07.2
US-17	Review the ingestion log and coverage update	Maintainer	EP-04	Must	2	R1	FR-07.3, NFR-07.2
US-18	Re-index amended minutes	Maintainer	EP-04	Should	3	R2	FR-07.5
US-19	Publish an index only if it passes the retrieval gate	Product owner	EP-05	Must	5	R1	NFR-01.1, NFR-01.2, NFR-01.3
US-20	Measure citation support and decline rate before release	Product owner	EP-05	Must	5	R1	NFR-02.1, NFR-02.2
US-21	Use the assistant with a keyboard and screen reader	P1 Resident	EP-06	Must	5	R1	NFR-05.1, NFR-05.2, NFR-05.3, NFR-05.4, NFR-05.5
US-22	Ask without an account or being tracked	P1 Resident	EP-06	Must	3	R1	NFR-06.1, NFR-06.2, NFR-06.3, NFR-06.4
US-23	Get an answer while the caller waits	P4 City staff	EP-06	Should	3	R2	NFR-03.1, NFR-03.2, NFR-03.3
US-24	Stay within the daily spending cap	Product owner	EP-06	Must	3	R1	NFR-04.1, NFR-04.2
US-25	Copy a citation into a reply	P4 City staff	EP-07	Should	2	R2	FR-02.1, FR-02.2
US-26	Report a wrong or unclear answer	P1 Resident	EP-07	Must	2	R1	TR-07, NFR-06.1

Totals: 26 stories, 90 points. Release 1 (MVP): 19 stories, 69 points. Release 2: 7 stories, 21 points.

5. Stories and acceptance criteria

EP-01 Ask and answer

US-01 Ask a question in plain language

As a resident, **I want** to ask about a council decision in my own words, **so that** I do not need to know the meeting date or the wording used in the minutes.

Priority	Estimate	Release	Satisfies	Verified by
Must	5 points	R1	FR-01.1, FR-01.2	UAT-01, UAT-04

```
Feature: Ask a question in plain language

  Scenario: Answer a question about a recorded decision
    Given the index includes the Regular Council Meeting minutes of 22 September 2026
    When I ask "Did council ratify the collective agreement with CUPE Local 1289?"
    Then I see an answer of 150 words or fewer
    And the answer says that council ratified the agreement
    And the answer cites the Regular Council Meeting of 22 September 2026

  Scenario: Question longer than the limit
    Given I am on the question page
    When I type a question longer than 500 characters
    Then I cannot submit it
    And I see a message that states the 500 character limit
```

US-02 Be told when the minutes do not contain the answer

As a resident, **I want** to be told plainly when the minutes do not answer my question, **so that** I do not rely on a guess and I know what to do next.

Priority	Estimate	Release	Satisfies	Verified by
Must	3 points	R1	FR-03.1, FR-03.3	UAT-06, UAT-07

```
Feature: Honest "not found" responses

  Scenario: Topic not recorded in the minutes
    Given a question that the UAT question bank marks as unanswerable
    When I ask that question
    Then I see that the answer was not found in the Regular Council Meeting minutes
    And the message names the coverage period that was searched
    And the message offers next steps: rephrase, browse the eSCRIBE portal, or contact the City Clerk's office
or 311
    And no citation is shown

  Scenario: Minutes not yet published
    Given the latest minutes in the index are from the meeting of 22 September 2026
    When I ask what council decided at the meeting of 6 October 2026
    Then I am told that minutes for that meeting are not yet published
    And I am told the latest meeting date covered
```

US-03 Decline questions outside council decisions

As a resident, **I want** the assistant to tell me when a question is not about recorded council decisions, **so that** I am not given opinions or predictions presented as facts.

Priority	Estimate	Release	Satisfies	Verified by
Must	2 points	R1	FR-03.2	UAT-08

```
Feature: Decline out-of-scope questions

  Scenario Outline: Opinions, predictions and legal advice are declined
    When I ask "<question>"
    Then I see a one-sentence explanation that the assistant only reports what the minutes record
    And no answer text or citation is shown

    Examples:
      | question                                          |
      | Should council approve more bike lanes?           |
      | Will my property taxes go up next year?           |
      | Is my neighbour's extension legal under the by-law? |
```

US-04 See all decisions on a topic in date order

As a community association volunteer, **I want** all decisions about a topic listed oldest to newest with their dates, **so that** I can see how a matter developed and which decision is the latest.

Priority	Estimate	Release	Satisfies	Verified by
Should	5 points	R2	FR-01.3	UAT-10

```
Feature: Decisions in date order

  Scenario: Topic decided at more than one meeting
    Given the evaluation set contains a topic decided at two or more Regular Council Meetings
```

```
    When I ask what council decided about that topic
    Then each decision is listed with its meeting date
    And the decisions appear oldest first
    And the answer says that a later decision may change an earlier one
```

US-05 Get the outcome exactly as recorded

As a journalist, **I want** the outcome reported exactly as the minutes record it, **so that** I do not report a deferral as an approval or an adoption in principle as a final adoption.

Priority	Estimate	Release	Satisfies	Verified by
Must	3 points	R1	FR-01.4	UAT-05

```
Feature: Outcomes reported as recorded

  Scenario: A deferral is not reported as an approval
    Given the minutes of 22 September 2026 record that council deferred the matter at 351–361 Kenmount Road for
two weeks
    When I ask "What did council decide about 351–361 Kenmount Road?"
    Then the answer says the matter was deferred for two weeks
    And the answer does not say the application was approved or rejected
    And the answer says that a later decision may exist after the coverage end date

  Scenario: An adoption in principle is not reported as a final adoption
    Given the minutes of 22 September 2026 record that council adopted two neighbourhood plans in principle
    When I ask "Did council adopt the Rennie's River and Central and Mundy Pond neighbourhood plans?"
    Then the answer says the plans were adopted in principle
    And the answer says staff were directed to prepare the related Municipal Plan and Development Regulations
amendments
    And the answer does not describe the plans as finally adopted
```

US-06 Understand where final decisions are made

As a resident, **I want** to be told, when I ask about Committee of the Whole, that final decisions are made at Regular Council Meetings, **so that** I look for the decision in the right place.

Priority	Estimate	Release	Satisfies	Verified by
Should	2 points	R2	FR-01.5	UAT-09

```
Feature: Committee of the Whole questions

  Scenario: Question about a Committee of the Whole meeting
    When I ask "What did Committee of the Whole decide on 15 September 2026?"
    Then I am told that Committee of the Whole makes no final decisions
    And I am told its report was considered at the Regular Council Meeting of 22 September 2026, item 8.1
    And that statement has a citation
```

EP-02 Trust and verification

US-07 See citations with meeting name, date and page

As a journalist, **I want** every statement in an answer cited with the meeting name, date and page, **so that** I can verify and quote it.

Priority	Estimate	Release	Satisfies	Verified by
Must	3 points	R1	FR-02.1, FR-02.4	UAT-02, UAT-04

```
Feature: Complete citations

  Scenario: Every statement is cited
    When I receive an answer that is not "not found"
    Then every statement in the answer has at least one numbered citation
    And each citation shows the meeting name, the meeting date and the page number
    And a citation reads like "Regular Council Meeting, 22 September 2026, page <n>"
```

US-08 Open the official document at the cited page

As a resident, **I want** a citation to open the official minutes at the cited page, **so that** I can check the answer myself.

Priority	Estimate	Release	Satisfies	Verified by
Must	2 points	R1	FR-02.2	UAT-03

```
Feature: Links to the official record

  Scenario: PDF citation opens the official document
    Given an answer cites PDF minutes on page <n>
    When I select the citation link
    Then the official minutes open from the City's eSCRIBE portal
    And the PDF viewer opens at page <n>, or the page number is shown next to the link if my browser does not
support page links
```

US-09 Cite HTML-only minutes by agenda item

As a journalist, **I want** answers based on HTML-only minutes to cite the agenda item, **so that** I can still find the exact place in the record.

Priority	Estimate	Release	Satisfies	Verified by
Should	3 points	R2	FR-02.3	UAT-12

```
Feature: Citations for HTML-only minutes

  Scenario: Answer from minutes published only in HTML
    Given the minutes of a meeting are published only in HTML
    When an answer cites those minutes
    Then the citation shows the meeting name, the meeting date and "item <number>"
    And the link opens the HTML minutes on the eSCRIBE portal
```

US-10 Read the source excerpts next to the answer

As a journalist, **I want** to read the source excerpts next to the answer, **so that** I can judge it without opening every document.

Priority	Estimate	Release	Satisfies	Verified by
Must	3 points	R1	FR-05.1, FR-05.2, FR-05.3	UAT-11, UAT-04

```
Feature: Source excerpts

  Scenario: Excerpts shown with the answer
    When I receive an answer that is not "not found"
    Then I see up to five excerpts next to the answer
    And each excerpt is 300 characters or fewer and copied word for word from the minutes
    And a shortened excerpt is marked with an ellipsis
    And each excerpt is labelled with the meeting name, date and page or item and links to the official
```

```
document
    And each numbered citation marker in the answer matches one excerpt
```

US-11 See coverage dates, disclaimer and model name

As a resident, **I want** to see what period the assistant covers, that it is not the official record, and which model produced the answer, **so that** I know how far to trust it.

Priority	Estimate	Release	Satisfies	Verified by
Must	2 points	R1	NFR-07.1, NFR-07.2, NFR-07.3, NFR-07.4	UAT-22

```
Feature: Transparency information

  Scenario: Information shown on the page
    When I open the assistant
    Then I see the coverage period with the first and latest meeting dates and the last update date
    And I see a disclaimer that the official minutes on eSCRIBE prevail
    And I see the name and version of the language model
    And I see the independence notice
```

EP-03 Find exact terms and filter

US-12 Find by exact identifier

As a community association volunteer, **I want** to search by an address, by-law number, resolution number or development file number, **so that** I find the exact item I am following.

Priority	Estimate	Release	Satisfies	Verified by
Must	5 points	R1	FR-06.1, FR-02.5	UAT-13, UAT-14

```
Feature: Exact identifiers

  Scenario Outline: An exact identifier retrieves the right passage
    When I ask about "<identifier>"
    Then the top five retrieved passages include the minutes of <meeting>, item <item>

    Examples:
      | identifier            | meeting           | item |
      | SJMC-R-2026-09-22/309 | 22 September 2026 | 15.1 |
      | DEV2600106            | 22 September 2026 | 6.4  |
      | 259 Blackmarsh Road   | 22 September 2026 | 6.8  |

  Scenario: Resolution number shown when present
    Given the cited passage contains the resolution number SJMC-R-2026-09-22/309
    When the answer cites that passage
    Then the resolution number is shown in the answer or the citation
```

US-13 Find with my own words

As a resident, **I want** to find decisions even when I use different words from the minutes, **so that** I do not have to guess the official wording.

Priority	Estimate	Release	Satisfies	Verified by
Must	5 points	R1	FR-06.2, FR-06.3	UAT-15

```
Feature: Paraphrased questions
```

```
  Scenario: Different words from the minutes
    When I ask "Did council approve the new contract with CUPE Local 1289?"
    Then the top five retrieved passages include the minutes of 22 September 2026, item 15.1
    And the answer cites that passage
```

FR-06.3 (how keyword and semantic results are combined) is a design requirement. It is verified through the subset reporting in Project A check PA-01 rather than by a user-facing scenario.

US-14 Limit the search to a date range

As a journalist, **I want** to limit the search to a date range, **so that** I only see decisions from the period I am reporting on.

Priority	Estimate	Release	Satisfies	Verified by
Should	3 points	R2	FR-04.1, FR-04.2, FR-04.3	UAT-16, UAT-17

```
Feature: Date range filter

  Scenario: The range excludes meetings outside it
    Given I set the date range from 1 January 2025 to 31 December 2025
    When I ask "Did council ratify the collective agreement with CUPE Local 1289?"
    Then no citation is from a meeting outside 2025
    And if no 2025 meeting answers the question, I am told it was not found in that period

  Scenario: Start date after end date
    When I set a start date that is later than the end date
    Then I see an error message that explains the problem in text
    And the question is not submitted until the range is corrected
```

EP-04 Keep the record current

US-15 Ingest newly published minutes on a schedule

As the maintainer, **I want** newly published Regular Council Meeting minutes ingested automatically on a schedule, **so that** answers stay current without manual editing.

Priority	Estimate	Release	Satisfies	Verified by
Must	8 points	R1	FR-07.1, FR-07.4	UAT-18

```
Feature: Scheduled ingestion

  Scenario: New minutes are published on the portal
    Given the index covers all minutes published up to the last run
    And minutes for a later Regular Council Meeting are then published on eSCRIBE
    When the scheduled ingestion runs
    Then the new minutes are indexed without any manual editing
    And questions about that meeting can be answered with citations

  Scenario: Polite access to the portal
    When the ingestion runs
    Then it sends at most one request per second with the descriptive User-Agent
    And it does not download documents that have not changed
    And it does not try to bypass any access control on the portal
```

Note: this is the largest story and carries the most uncertainty because of constraint C-03 (browser verification on PDF requests). If the access method is not agreed before sprint planning, split it into (a) detect newly listed minutes and ingest HTML minutes, and (b) ingest PDF minutes once an access method is agreed (OQ-05).

US-16 Re-run ingestion without duplicates

As the maintainer, **I want** to re-run ingestion without creating duplicates, **so that** I can recover from a failed run safely.

Priority	Estimate	Release	Satisfies	Verified by
Must	3 points	R1	FR-07.2	UAT-19

```
Feature: Idempotent ingestion

  Scenario: Re-run with nothing new
    Given an ingestion run has completed
    When I run ingestion again and no minutes are new or changed
    Then the numbers of meetings, documents and passages in the index are unchanged
```

US-17 Review the ingestion log and coverage update

As the maintainer, **I want** an ingestion log and updated coverage dates after every run, **so that** I can see what changed and users can see how current the data is.

Priority	Estimate	Release	Satisfies	Verified by
Must	2 points	R1	FR-07.3, NFR-07.2	UAT-20

```
Feature: Ingestion log and coverage

  Scenario: After a run
    When an ingestion run finishes
    Then the log records the meetings found, added, unchanged and failed
    And the coverage period shown to users matches the meetings in the index
    And the last update date is the date of the run
```

US-18 Re-index amended minutes

As the maintainer, **I want** changed minutes detected and re-indexed, **so that** answers reflect corrected documents.

Priority	Estimate	Release	Satisfies	Verified by
Should	3 points	R2	FR-07.5	UAT-19

```
Feature: Amended minutes

  Scenario: A document's content changes
    Given a minutes document in the index is replaced on the portal with changed content
    When ingestion runs
    Then the changed document is re-indexed
    And passages from the earlier version are no longer returned
```

EP-05 Quality gates

US-19 Publish an index only if it passes the retrieval gate

As the product owner, **I want** a new index published only if it passes the retrieval quality gate, **so that** a faulty ingestion cannot lower answer quality.

Priority	Estimate	Release	Satisfies	Verified by
Must	5 points	R1	NFR-01.1, NFR-01.2, NFR-01.3	UAT-21

```
Feature: Retrieval quality gate

  Scenario: The gate passes
    Given a new index has been built
    When the evaluation suite reports hit@5 of at least 85% on the answerable questions
    Then the new index is published
    And the coverage dates are updated

  Scenario: The gate fails
    Given a new index has been built
    When the evaluation suite reports hit@5 below 85%
    Then the previous index stays live
    And the maintainer is alerted with the evaluation report
```

US-20 Measure citation support and decline rate before release

As the product owner, **I want** citation support and the decline rate measured before each release, **so that** I can show the answers meet the agreed quality thresholds.

Priority	Estimate	Release	Satisfies	Verified by
Must	5 points	R1	NFR-02.1, NFR-02.2	UAT-04

```
Feature: Answer quality report

  Scenario: Report before release
    When the answer evaluation runs on the current evaluation set
    Then the report shows the share of citations that support their claim
    And the share of unanswerable questions that were declined
    And the evaluation set version, the model name and the date
    And the release is blocked if citation support is below 90% or the decline rate is below 80%
```

EP-06 Accessible, private and affordable

US-21 Use the assistant with a keyboard and screen reader

As a resident who uses a screen reader, **I want** to ask questions and hear the answers and citations announced, **so that** I can use the service independently.

Priority	Estimate	Release	Satisfies	Verified by
Must	5 points	R1	NFR-05.1, NFR-05.2, NFR-05.3, NFR-05.4, NFR-05.5	UAT-23, UAT-24, UAT-25

```
Feature: Accessible use

  Scenario: Keyboard only
    Given I use only a keyboard
    When I move through the page, ask a question and open a citation
    Then every control is reachable in a logical order
    And the focus is always visible

  Scenario: Screen reader
    Given I use a screen reader
    When an answer is ready
    Then the answer is announced without my focus moving unexpectedly
    And the question box and the date fields have spoken labels

  Scenario: Zoom and contrast
    When I zoom the page to 200% or narrow it to 320 pixels wide
```

```
    Then no content or function is lost and there is no horizontal scrolling for text
    And text meets a contrast ratio of at least 4.5 to 1
```

US-22 Ask without an account or being tracked

As a resident, **I want** to use the assistant without an account and without being tracked, **so that** my questions cannot be linked to me.

Priority	Estimate	Release	Satisfies	Verified by
Must	3 points	R1	NFR-06.1, NFR-06.2, NFR-06.3, NFR-06.4	UAT-26

```
Feature: Privacy by design

  Scenario: No account and no tracking
    When I use the assistant
    Then I am never asked to sign in or give personal information
    And no tracking cookies or third-party analytics are set
    And I see a privacy notice near the question box

  Scenario: Logs contain no identifiers
    Given I have asked a question
    When the maintainer inspects the application logs
    Then the log entry contains the question text, the date and quality data only
    And it contains no IP address or other identifier
    And entries older than the retention period have been deleted
```

US-23 Get an answer while the caller waits

As a City staff member answering an enquiry, **I want** answers within a few seconds, **so that** I can respond while the caller waits.

Priority	Estimate	Release	Satisfies	Verified by
Should	3 points	R2	NFR-03.1, NFR-03.2, NFR-03.3	UAT-27

```
Feature: Response time

  Scenario: Performance test
    When the performance test runs over the evaluation set
    Then the 95th percentile end-to-end answer time is under 10 seconds
    And search-only results appear in under 1 second after the page has loaded

  Scenario: Progress indicator
    When I ask a question
    Then a progress indicator is shown until the answer appears
    And the status is announced to assistive technology
```

US-24 Stay within the daily spending cap

As the product owner, **I want** a daily spending cap with an automatic switch to search-only mode, **so that** running costs stay within budget.

Priority	Estimate	Release	Satisfies	Verified by
Must	3 points	R1	NFR-04.1, NFR-04.2	UAT-28

```
Feature: Spending cap

  Scenario: The daily cap is reached
    Given the daily spending cap has been reached
```

```
    When a user asks a question
    Then the assistant returns search results with citations but no generated answer
    And a notice explains that generated answers are temporarily unavailable

  Scenario: Cost report
    When the cost report runs
    Then it shows the average model cost per answer in CAD calculated from the provider's usage records
    And it shows the exchange rate and date used
```

EP-07 Staff support and feedback

US-25 Copy a citation into a reply

As a City staff member answering an enquiry, **I want** to copy a citation and its official link in one step, **so that** I can paste a reference the resident can check into my reply.

Priority	Estimate	Release	Satisfies	Verified by
Should	2 points	R2	FR-02.1, FR-02.2	UAT-29

```
Feature: Copy citation

  Scenario: Copy a citation as plain text
    Given an answer with citations is shown
    When I choose "Copy citation" for one citation
    Then the meeting name, meeting date, page or item and the official link are copied as plain text
    And a confirmation is announced to assistive technology
```

US-26 Report a wrong or unclear answer

As a resident, **I want** to report an answer that looks wrong without giving personal information, **so that** it can be corrected for everyone.

Priority	Estimate	Release	Satisfies	Verified by
Must	2 points	R1	TR-07, NFR-06.1	UAT-30

```
Feature: Report a problem

  Scenario: Report an answer
    Given an answer is shown
    When I choose "Report a problem" and describe the issue
    Then the report is sent with the question, the answer and its citations
    And I am not asked for my name or contact details
    And the report appears in the maintainer's triage queue
```

6. INVEST review

Criterion	How the backlog meets it	Exceptions and actions
Independent	Stories are split by capability, not by technical layer. Dependencies are sequencing only: ingestion (US-15) comes before answering (US-01), and US-25 builds on US-07.	None that share scope.
Negotiable	Stories state needs and outcomes, not screen designs. For example, US-25 could be a button or a keyboard shortcut.	Wording of the "not found" and disclaimer messages depends on OQ-07.
Valuable	Every story names who benefits and why. Enabler stories (US-16, US-19, US-20, US-24) are written for the product owner or maintainer because they protect quality, cost and recovery.	None.

Criterion	How the backlog meets it	Exceptions and actions
Estimable	Acceptance criteria and test data exist for every story.	US-15 is the least certain because of C-03; see the split plan in its note.
Small	The largest story is 8 points and fits in one sprint at the planning velocity below.	US-15 is the split candidate.
Testable	Every story has Gherkin scenarios and at least one UAT case, and the measurable ones also have a Project A automated check.	Coverage is checked automatically (see 04-traceability-matrix.md).

7. Prioritization rationale

1. **Trust before convenience.** Everything BR-02 depends on is Must: citations, "not found", excerpts, transparency, quality gates and the problem-report channel. An AI answer that cannot be checked fails the purpose and creates risk (R-01, R-03).
2. **The core finding capability is Must.** Plain-language questions (US-01) and hybrid retrieval for exact terms and paraphrases (US-12, US-13) deliver BR-01.
3. **Currency is Must.** Without scheduled, safe ingestion (US-15 to US-17), answers go stale quickly (R-02), because minutes are published after adoption at a later meeting.
4. **Public-sector obligations are Must.** Accessibility (US-21), privacy (US-22) and the spending cap (US-24) are conditions for any public release, not enhancements.
5. **Should stories improve the experience or serve narrower needs, and have workarounds.** People can add the year to a question until the date filter (US-14) exists. HTML-only minutes still get a meeting-level citation until US-09 adds item numbers, and they affect 9 of 242 meetings. Staff can copy text by hand until US-25. Response-time tuning (US-23) matters most for staff on the phone.
6. **Later and Won't.** Structured motion data (resolution numbers and vote counts as fields), Committee of the Whole minutes and decision notes are later-release candidates that depend on OQ-06, OQ-13 and OQ-14. Accounts, saved histories, alerts, agendas, video and other languages are out of scope for now (BRD section 4.2).

8. Release plan

Planning velocity is assumed at 20 points per two-week sprint for a two-person team, to be replaced by measured velocity after the first sprint.

Release	Sprint	Stories	Points	Outcome
R1 (MVP)	Sprint 1	US-15, US-16, US-17, US-19	18	Minutes ingested safely; retrieval gate and evaluation set in place
R1 (MVP)	Sprint 2	US-12, US-13, US-01, US-03, US-08	19	Questions answered from hybrid retrieval, with links to the official record
R1 (MVP)	Sprint 3	US-02, US-05, US-07, US-10, US-11, US-20	19	Full citations, excerpts, honest "not found", answer quality gate
R1 (MVP)	Sprint 4	US-21, US-22, US-24, US-26	13	Accessibility, privacy, cost cap and problem reports; remaining capacity for UAT fixes
R1		**Release 1 total**	**69**	UAT and pilot (TR-03) start after Sprint 4
R2	Sprint 5 and part of Sprint 6	US-04, US-06, US-09, US-14, US-18, US-23, US-25	21	Date filter, decisions in date order, HTML item citations, staff copy, amended minutes, performance tuning
Later	To be planned	Structured motions, Committee of the Whole minutes, decision notes	Not estimated	Subject to OQ-06, OQ-13, OQ-14

MVP definition: a member of the public can ask a question about Regular Council Meeting minutes and get either a short answer with complete citations, verbatim excerpts and links to the official record, or an honest "not found". The service is accessible, collects no personal data, keeps costs capped, and only publishes indexes that pass the quality gates.

User Acceptance Test Plan: St. John's Council Minutes Assistant

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Item	Value
Document ID	CMA-BA-03
Version	0.9.1 (draft for review; Project A results recorded 18 October 2026)
Author	Mostafa Rafiur Wasib (UAT lead)
Date	17 October 2026
Tests	Release 1 (MVP) and Release 2 stories in 02-user-stories.md
Requirements	01-business-requirements-document.md
Traceability	04-traceability-matrix.md and traceability.csv
Execution status	UAT not started: all 30 test cases are "Not run". Project A's automated check results were recorded on 18 October 2026 (section 14): 13 of the 14 checks met their thresholds, and PA-05 (NFR-03 answer time) did not.

1. Objectives

1. Confirm with representatives of each persona that the assistant meets the business requirements BR-01 to BR-03 in realistic use.
2. Confirm that every Must requirement in the BRD is met and that each Should requirement is met or has an agreed deferral.
3. Confirm the AI-specific safeguards work: answers are supported by citations, outcomes are reported as recorded, and the assistant declines rather than guesses (BR-02, R-01 to R-04).
4. Confirm accessibility (WCAG 2.1 AA), privacy and transparency behaviour with the people who will judge them: assistive technology users and the ATIPP Coordinator.
5. Gather the evidence needed for the go/no-go decision (TR-09).

2. Scope

In scope: stories US-01 to US-26; the requirements they satisfy; the measured quality gates from Project A (section 14); the support process for problem reports (TR-07).

Out of scope: unit, integration and system testing (owned by the delivery team in Project A); load testing beyond the NFR-03 measurement; security penetration testing (recommended before any production use); the content of the City's official minutes.

3. Approach

- **Business acceptance, not technical testing.** Testers act as their persona and judge whether the result is fit for purpose. The delivery team has already completed system testing (entry criterion 2).

- **Risk-based priority.** Test case priority follows the MoSCoW priority of the requirement: Must is High, Should is Medium, Could is Low. AI-risk cases (UAT-04, UAT-05, UAT-08) are run first because their failures are the most harmful.
- **Manual and automated evidence together.** Measurable thresholds (hit@5, citation support, decline rate, latency, cost) are verified by Project A's automated evaluation suite on the UAT index (section 14). UAT testers check the same behaviour by sampling and judging real answers.
- **Repeat AI judgements.** Model output can vary between runs. Each test case that judges an answer is run twice; a failure in either run is logged. The model version and index snapshot are recorded with every result.
- **Two cycles.** Cycle 1 runs every test case. Cycle 2 retests fixed defects and re-runs all High priority cases and the full automated suite as regression.
- **Evidence.** For each executed step, the tester records the result, a screenshot or log extract, the date, the model version and the index snapshot ID. Screenshots must not contain personal information.

4. Entry criteria

1. The BRD is baselined (version 1.0) and the stories in scope meet the Definition of Done.
2. System and integration testing are complete, with no open Severity 1 or Severity 2 defects.
3. Project A automated checks (section 14) have run on the UAT index snapshot, and the NFR-01 and NFR-02 quality gates are met.
4. The UAT environment is available with a frozen index snapshot whose coverage period is recorded.
5. Expected results in the question bank (section 9) have been confirmed against the official minutes, including page numbers.
6. Testers are recruited and briefed, and accessibility testers have their usual assistive technology ready.

5. Exit criteria

1. All High priority test cases have been executed, and at least 90% of all test cases have been executed.
2. All High priority test cases have passed, at first run or after retest.
3. No Severity 1 or Severity 2 defects are open.
4. No more than five Severity 3 defects are open, each with an agreed workaround and fix date.
5. On the release candidate, Project A reports hit@5 of at least 85% (NFR-01), citation support of at least 90% and a decline rate of at least 80% (NFR-02), and an average model cost per answer under CAD $0.05 (NFR-04). The NFR-03 response times are measured and reported.
6. No open WCAG 2.1 AA failures from UAT-23 to UAT-25.
7. The UAT summary report is issued, and the business sponsor and product owner have signed off (section 15).

6. Roles and responsibilities

Role	Responsibilities	Assigned to
UAT lead	Plans and coordinates UAT, prepares data, runs triage, writes the summary report	Business analyst
Business sponsor	Decides on exit and signs off	City Clerk's office (proposed)
Product owner	Prioritizes defects, accepts results, decides on deferrals	To be confirmed

Role	Responsibilities	Assigned to
Testers	Execute test cases as their persona and record evidence	Two residents, one journalist or researcher, one community association volunteer, two City staff who answer enquiries (to recruit)
Accessibility tester	Executes UAT-23 to UAT-25 with a screen reader and magnification	At least one person who uses assistive technology daily (to recruit)
Privacy reviewer	Reviews UAT-26 and UAT-30 evidence	ATIPP Coordinator
Developer support	Fixes defects, re-runs automated checks, supports ingestion tests	Project A developer

7. Test environment

Item	Specification
Application	UAT instance of the assistant, separate from the public demo; address to be confirmed
Index	Frozen snapshot; snapshot ID and coverage period recorded at the start of UAT
Ingestion tests	A separate ingestion test environment. UAT-18 starts from an index that excludes the most recently published meeting, so the test does not depend on new minutes being published during UAT.
Language model	Name and version pinned for the UAT period
Desktop browsers	Current versions of Chrome, Edge, Firefox and Safari
Mobile	Safari on iOS and Chrome on Android
Assistive technology	NVDA with Firefox or Chrome on Windows; VoiceOver with Safari on macOS and iOS; browser zoom to 200%
Accounts	None needed (NFR-06.1)

8. Test data

- **UAT question bank** (section 9): questions with expected results taken from published minutes. Expected results apply to the frozen snapshot. If the snapshot includes minutes published after 17 October 2026, the expected results are re-checked during preparation; for example, a later meeting may have decided the deferred matter in Q-02.
- **Project A evaluation set:** answerable and unanswerable questions with known sources (NFR-01.1). Its version is recorded in each report. UAT-06 uses unanswerable questions from this set, because each has been checked against the index.
- **Long input:** a 501-character question for UAT-01.
- **Accessibility checklist:** the WCAG 2.1 AA success criteria named in NFR-05.
- **No personal information** is used in any test question.

9. Sample UAT question bank

Facts below were checked against the HTML minutes of the Regular Council Meeting of 22 September 2026 and the portal listing on 17 October 2026. Page numbers are confirmed from the official PDF during preparation.

ID	Question	Type	Expected result	Used in
Q-01	Did council ratify the collective agreement with CUPE Local 1289?	Answerable	Yes. Regular Council Meeting, 22 September 2026, item 15.1; motion carried 8 to 2; resolution SJMC-R-2026-09-22/309.	UAT-01 to UAT-04, UAT-11

ID	Question	Type	Expected result	Used in
Q-02	What did council decide about 351-361 Kenmount Road?	Answerable, nuanced outcome	Council deferred the matter for two weeks (22 September 2026, item 6.4; carried 8 to 1, with one abstention after a declared conflict of interest). Must not be reported as approved; must note that a later decision may exist.	UAT-05
Q-03	Did council adopt the Rennie's River and Central and Mundy Pond neighbourhood plans?	Answerable, nuanced outcome	Adopted in principle, with staff directed to prepare the related Municipal Plan and Development Regulations amendments (22 September 2026, item 8.1; carried 10 to 0; resolution SJMC-R-2026-09-22/306). Must not be reported as finally adopted.	UAT-05
Q-04	SJMC-R-2026-09-22/309	Exact identifier	Same passage as Q-01.	UAT-13
Q-05	DEV2600106; and separately, 259 Blackmarsh Road	Exact identifier	22 September 2026, item 6.4; and 22 September 2026, item 6.8.	UAT-14
Q-06	Did council approve the new contract with CUPE Local 1289?	Paraphrase	Same passage as Q-01.	UAT-15
Q-07	What did council decide at the meeting of 6 October 2026?	Minutes not yet published	At the snapshot date the minutes were not listed. Expected: "not yet published", with the latest meeting covered (22 September 2026).	UAT-07
Q-08	What did Committee of the Whole decide on 15 September 2026?	Scope explanation	Committee of the Whole makes no final decisions. Its report was considered at the Regular Council Meeting of 22 September 2026, item 8.1.	UAT-09
Q-09	Should council approve more bike lanes?	Out of scope	Declined with a one-sentence explanation.	UAT-08
Q-10	Q-01 with the date range 1 January 2025 to 31 December 2025	Date filter	No citation outside 2025; not found in that period.	UAT-16

10. Defect management

Severity (impact on the business)

Severity	Definition	Examples
S1 Critical	Wrong or harmful output presented as fact, a privacy or security breach, or the service is unusable. No workaround.	An answer contradicts its cited passage; a citation points to a passage that does not exist; questions stored with an IP address; outage.
S2 Major	A Must requirement fails, or a main function is impaired with no reasonable workaround.	A citation link opens the wrong meeting; a keyboard trap; the assistant guesses instead of saying "not found"; the spending cap is not enforced.
S3 Minor	A function works incorrectly but a workaround exists, or a Should requirement fails.	An unclear date error message; a missing ellipsis on a shortened excerpt; page anchors not supported in one browser.
S4 Cosmetic	Presentation issue with no effect on function or meaning.	Spacing; a typing error in help text.

Priority (urgency of the fix)

Priority	Definition
P1 Immediate	Blocks further testing or go-live. Fix now.
P2 High	Fix before go-live.
P3 Medium	Fix in the next release or patch; workaround documented.
P4 Low	Add to the backlog.

Lifecycle: New, Triaged, In progress, Fixed, Ready for retest, Closed (or Reopened). A defect may be Deferred only with the product owner's written approval and a recorded reason.

Triage: daily during execution, with the UAT lead, product owner and developer. Any S1 is escalated to the business sponsor the same day. Any answer-quality defect (S1 or S2) is added as a regression item to the Project A evaluation set.

11. Schedule template

Dates are set when the UAT environment is confirmed. Durations are suggestions.

Phase	Activities	Owner	Suggested duration	Planned start	Planned end	Status
Preparation	Confirm question bank pages; set up environment and snapshot; recruit testers	UAT lead	1 week	TBD	TBD	Not started
Briefing	30-minute tester briefing and persona scripts	UAT lead	1 day	TBD	TBD	Not started
Cycle 1	Execute all test cases; run the automated suite on the snapshot	Testers, developer	1 week	TBD	TBD	Not started
Fixing	Fix S1 and S2 defects; agree S3 workarounds	Developer, product owner	1 week	TBD	TBD	Not started
Cycle 2	Retest; regression of High priority cases; re-run the automated suite	Testers, developer	3 days	TBD	TBD	Not started
Exit review	Summary report, go/no-go recommendation, sign-off	UAT lead, sponsor, product owner	2 days	TBD	TBD	Not started

12. Test case summary

ID	Test case	Story	Requirements	Type	Priority	Status
UAT-01	Answer a plain-language question with citations	US-01	FR-01.1, FR-01.2	Functional	High	Not run
UAT-02	Citation shows meeting name, date and page	US-07	FR-02.1	Functional	High	Not run
UAT-03	Citation link opens the official document at the cited page	US-08	FR-02.2	Functional	High	Not run
UAT-04	Every statement is supported by its cited passage	US-01, US-07, US-10, US-20	FR-01.2, FR-02.4, NFR-02.1	AI quality	High	Not run
UAT-05	Outcome reported exactly as recorded	US-05	FR-01.4	AI quality	High	Not run
UAT-06	"Not found" response for an unanswerable question	US-02	FR-03.1	Functional	High	Not run
UAT-07	Minutes not yet published	US-02	FR-03.3	Functional	Medium	Not run
UAT-08	Out-of-scope questions declined	US-03	FR-03.2	AI quality	High	Not run
UAT-09	Committee of the Whole question explained	US-06	FR-01.5	Functional	Medium	Not run
UAT-10	Several decisions listed in date order	US-04	FR-01.3	Functional	Medium	Not run
UAT-11	Source excerpts shown with the answer	US-10	FR-05.1, FR-05.2, FR-05.3	Functional	High	Not run

ID	Test case	Story	Requirements	Type	Priority	Status
UAT-12	HTML-only minutes cited by agenda item	US-09	FR-02.3	Functional	Medium	Not run
UAT-13	Resolution number retrieves the right passage	US-12	FR-06.1, FR-02.5	Functional	High	Not run
UAT-14	File number and civic address retrieve the right passage	US-12	FR-06.1	Functional	High	Not run
UAT-15	Paraphrased question retrieves the right passage	US-13	FR-06.2	Functional	High	Not run
UAT-16	Date range restricts the sources	US-14	FR-04.1, FR-04.3	Functional	Medium	Not run
UAT-17	Invalid date range rejected with a clear message	US-14	FR-04.2	Functional	Medium	Not run
UAT-18	New minutes ingested on schedule without manual editing	US-15	FR-07.1, FR-07.4	Operational	High	Not run
UAT-19	Re-run creates no duplicates; changed minutes re-indexed	US-16, US-18	FR-07.2, FR-07.5	Operational	High	Not run
UAT-20	Ingestion log and coverage dates updated	US-17	FR-07.3, NFR-07.2	Operational	High	Not run
UAT-21	Failed quality gate keeps the previous index	US-19	NFR-01.2	Operational	High	Not run
UAT-22	Disclaimer, coverage, model name and independence notice shown	US-11	NFR-07.1, NFR-07.2, NFR-07.3, NFR-07.4	Transparency	High	Not run
UAT-23	Keyboard-only use	US-21	NFR-05.1	Accessibility	High	Not run
UAT-24	Screen reader labels and announcements	US-21	NFR-05.3, NFR-05.4	Accessibility	High	Not run
UAT-25	Contrast, zoom and reflow	US-21	NFR-05.2, NFR-05.5	Accessibility	High	Not run
UAT-26	No account, no tracking, no identifiers in logs	US-22	NFR-06.1, NFR-06.2, NFR-06.3, NFR-06.4	Privacy	High	Not run
UAT-27	Response time and progress indicator	US-23	NFR-03.1, NFR-03.2, NFR-03.3	Performance	Medium	Not run
UAT-28	Spending cap switches to search-only mode	US-24	NFR-04.1	Operational	High	Not run
UAT-29	Staff copies a citation into a reply	US-25	FR-02.1, FR-02.2	Functional	Medium	Not run
UAT-30	Report a wrong answer without personal information	US-26	TR-07, NFR-06.1	Support	High	Not run

Totals: 30 test cases: 22 High priority and 8 Medium priority.

13. Test cases

UAT-01 Answer a plain-language question with citations

Story: US-01. **Requirements:** FR-01.1, FR-01.2. **Priority:** High. **Status:** Not run.

Preconditions: UAT snapshot includes the minutes of 22 September 2026. Tester acts as P1 Resident.

Steps

1. Open the assistant.
2. Ask Q-01: "Did council ratify the collective agreement with CUPE Local 1289?"
3. Count the words in the answer.
4. Paste a 501-character question and try to submit it.

Expected result

- The answer says council ratified the agreement, is 150 words or fewer, and cites the Regular Council Meeting of 22 September 2026.
- The 501-character question cannot be submitted, and the 500 character limit is stated.

UAT-02 Citation shows meeting name, date and page

Story: US-07. **Requirements:** FR-02.1. **Priority:** High. **Status:** Not run.

Preconditions: UAT-01 answer available.

Steps

1. Ask Q-01.
2. Read each citation.

Expected result

- Each citation shows the meeting name, meeting date and page, for example "Regular Council Meeting, 22 September 2026, page n", where n matches the page confirmed in preparation.

UAT-03 Citation link opens the official document at the cited page

Story: US-08. **Requirements:** FR-02.2. **Priority:** High. **Status:** Not run.

Preconditions: Q-01 answer available. Test in Chrome, Edge, Firefox and Safari.

Steps

1. Select the first citation link.
2. Note the address and the page shown.
3. Repeat in each desktop browser.

Expected result

- The official minutes for 22 September 2026 open from the City's eSCRIBE portal.
- The PDF opens at the cited page, or, where the browser does not support page links, the page number is shown next to the link.

UAT-04 Every statement is supported by its cited passage

Story: US-01, US-07, US-10, US-20. **Requirements:** FR-01.2, FR-02.4, NFR-02.1. **Priority:** High. **Status:** Not run.

Preconditions: A sample of 10 answerable questions from the evaluation set, plus Q-01 to Q-03. Two reviewers.

Steps

1. Ask each question twice (two runs).
2. For every statement in each answer, open the cited passage and judge whether it supports the statement.
3. Record any statement with no citation, and any citation that does not support its statement.
4. Compare the reviewers' judgements with the Project A citation support result (PA-03).

Expected result

- No statement is uncited.
- No citation contradicts its statement; at least 90% of citations support their statements.
- The reviewers' judgements broadly agree with PA-03; any large difference is logged for the delivery team.

UAT-05 Outcome reported exactly as recorded

Story: US-05. **Requirements:** FR-01.4. **Priority:** High. **Status:** Not run.

Preconditions: Snapshot includes the minutes of 22 September 2026. Tester acts as P2 Journalist.

Steps

1. Ask Q-02: "What did council decide about 351-361 Kenmount Road?"
2. Ask Q-03: "Did council adopt the Rennie's River and Central and Mundy Pond neighbourhood plans?"
3. Repeat both questions in a second run.

Expected result

- Q-02: the answer says the matter was deferred for two weeks, does not describe it as approved or rejected, and notes that a later decision may exist after the coverage end date.
- Q-03: the answer says the plans were adopted in principle and that staff were directed to prepare the related amendments, and does not describe the plans as finally adopted.

UAT-06 "Not found" response for an unanswerable question

Story: US-02. **Requirements:** FR-03.1. **Priority:** High. **Status:** Not run.

Preconditions: Three unanswerable questions from the Project A evaluation set (IDs recorded).

Steps

1. Ask each unanswerable question.
2. Read the response.

Expected result

- Each response says the answer was not found in the Regular Council Meeting minutes, names the coverage period searched, and offers next steps (rephrase, browse eSCRIBE, contact the City Clerk's office or 311).
- No answer text or citation is shown.

UAT-07 Minutes not yet published

Story: US-02. **Requirements:** FR-03.3. **Priority:** Medium. **Status:** Not run.

Preconditions: Snapshot's latest meeting with minutes is known (22 September 2026 at the 17 October 2026 snapshot).

Steps

1. Ask Q-07: "What did council decide at the meeting of 6 October 2026?"

Expected result

- The response says the minutes for that meeting are not yet published and gives the latest meeting date covered.

UAT-08 Out-of-scope questions declined

Story: US-03. **Requirements:** FR-03.2. **Priority:** High. **Status:** Not run.

Preconditions: None.

Steps

1. Ask Q-09: "Should council approve more bike lanes?"
2. Ask "Will my property taxes go up next year?"
3. Ask "Is my neighbour's extension legal under the by-law?"

Expected result

- Each is declined with a one-sentence explanation that the assistant only reports what the minutes record.
- No opinion, prediction or legal interpretation is given, and no citation is shown.

UAT-09 Committee of the Whole question explained

Story: US-06. **Requirements:** FR-01.5. **Priority:** Medium. **Status:** Not run.

Preconditions: Snapshot includes the minutes of 22 September 2026.

Steps

1. Ask Q-08: "What did Committee of the Whole decide on 15 September 2026?"

Expected result

- The response explains that Committee of the Whole makes no final decisions.
- It states, with a citation, that the committee's report was considered at the Regular Council Meeting of 22 September 2026, item 8.1.

UAT-10 Several decisions listed in date order

Story: US-04. **Requirements:** FR-01.3. **Priority:** Medium. **Status:** Not run.

Preconditions: An evaluation set topic decided at two or more Regular Council Meetings (ID recorded).

Steps

1. Ask what council decided about that topic.

Expected result

- Each decision is listed with its meeting date, oldest first, and each has a citation.
- The answer says that a later decision may change an earlier one.

UAT-11 Source excerpts shown with the answer

Story: US-10. **Requirements:** FR-05.1, FR-05.2, FR-05.3. **Priority:** High. **Status:** Not run.

Preconditions: Q-01 answer available.

Steps

1. Ask Q-01.
2. Count the excerpts and the characters in each.
3. Compare each excerpt with the official minutes.
4. Match each numbered citation marker to its excerpt.

Expected result

- Up to five excerpts, each 300 characters or fewer, word for word as in the minutes, with an ellipsis where shortened.

- Each excerpt is labelled with meeting name, date and page and links to the official document.
- Each numbered marker in the answer matches one excerpt.

UAT-12 HTML-only minutes cited by agenda item

Story: US-09. **Requirements:** FR-02.3. **Priority:** Medium. **Status:** Not run.

Preconditions: A question whose answer is in one of the nine meetings with HTML-only minutes (for example, the meeting of 29 July 2025), with expected item confirmed in preparation.

Steps

1. Ask the prepared question.
2. Read the citation and open its link.

Expected result

- The citation shows the meeting name, meeting date and "item n".
- The link opens the HTML minutes on the eSCRIBE portal.

UAT-13 Resolution number retrieves the right passage

Story: US-12. **Requirements:** FR-06.1, FR-02.5. **Priority:** High. **Status:** Not run.

Preconditions: Snapshot includes the minutes of 22 September 2026.

Steps

1. Ask Q-04: "SJMC-R-2026-09-22/309".
2. Review the retrieved passages and the answer.

Expected result

- The passage for item 15.1 of 22 September 2026 is among the top five results.
- The answer or citation shows the resolution number.

UAT-14 File number and civic address retrieve the right passage

Story: US-12. **Requirements:** FR-06.1. **Priority:** High. **Status:** Not run.

Preconditions: Snapshot includes the minutes of 22 September 2026. Tester acts as P3 Community volunteer.

Steps

1. Ask "DEV2600106".
2. Ask "What did council decide about 259 Blackmarsh Road?"

Expected result

- Step 1: item 6.4 of 22 September 2026 is among the top five results.
- Step 2: item 6.8 of 22 September 2026 is among the top five results and is cited in the answer.

UAT-15 Paraphrased question retrieves the right passage

Story: US-13. **Requirements:** FR-06.2. **Priority:** High. **Status:** Not run.

Preconditions: Snapshot includes the minutes of 22 September 2026.

Steps

1. Ask Q-06: "Did council approve the new contract with CUPE Local 1289?"

Expected result

- Item 15.1 of 22 September 2026 is among the top five results and is cited, even though the minutes use "ratify" and "collective agreement".

UAT-16 Date range restricts the sources

Story: US-14. **Requirements:** FR-04.1, FR-04.3. **Priority:** Medium. **Status:** Not run.

Preconditions: None.

Steps

1. Leave the dates blank and ask Q-01; note the citations.
2. Set the range 1 January 2025 to 31 December 2025 and ask Q-01 again (Q-10).

Expected result

- Step 1 searches the full coverage period and cites 22 September 2026.
- Step 2 cites no meeting outside 2025 and reports that the answer was not found in that period.

UAT-17 Invalid date range rejected with a clear message

Story: US-14. **Requirements:** FR-04.2. **Priority:** Medium. **Status:** Not run.

Preconditions: None. Run once with a keyboard and once with a screen reader.

Steps

1. Set the start date to 31 December 2025 and the end date to 1 January 2025.
2. Try to submit a question.

Expected result

- The question is not submitted.
- An error message explains in text that the start date must be on or before the end date, and the screen reader announces it.

UAT-18 New minutes ingested on schedule without manual editing

Story: US-15. **Requirements:** FR-07.1, FR-07.4. **Priority:** High. **Status:** Not run.

Preconditions: Ingestion test environment with an index that excludes the most recently published Regular Council Meeting. Request logging enabled.

Steps

1. Ask a question answered only by the excluded meeting; confirm "not found".
2. Let the scheduled ingestion run (or trigger the scheduled job).
3. Ask the same question again.
4. Review the request log for the run.

Expected result

- After the run, the question is answered with a citation to the newly ingested meeting, with no manual editing.
- The log shows at most one request per second, the descriptive User-Agent, no re-download of unchanged documents, and no attempt to bypass access controls.

UAT-19 Re-run creates no duplicates; changed minutes re-indexed

Story: US-16, US-18. **Requirements:** FR-07.2, FR-07.5. **Priority:** High. **Status:** Not run.

Preconditions: Ingestion test environment after UAT-18. A test copy of one document with changed content is available to the pipeline.

Steps

1. Record the counts of meetings, documents and passages.
2. Run ingestion again with nothing new; record the counts.
3. Present the changed document version; run ingestion.
4. Ask a question about the changed text.

Expected result

- Step 2 counts equal step 1 counts.
- After step 3 the changed document is re-indexed, and passages from the earlier version are no longer returned.

UAT-20 Ingestion log and coverage dates updated

Story: US-17. **Requirements:** FR-07.3, NFR-07.2. **Priority:** High. **Status:** Not run.

Preconditions: UAT-18 completed.

Steps

1. Open the ingestion log for the run.
2. Open the assistant and read the coverage period and last update date.

Expected result

- The log lists meetings found, added, unchanged and failed.
- The coverage period matches the meetings in the index, and the last update date is the date of the run.

UAT-21 Failed quality gate keeps the previous index

Story: US-19. **Requirements:** NFR-01.2. **Priority:** High. **Status:** Not run.

Preconditions: Ingestion test environment. A deliberately degraded test index (for example, built with a corrupted passage set) that scores below 85% hit@5.

Steps

1. Run the publish step with the degraded index.
2. Ask Q-01 in the assistant.
3. Check the maintainer's alerts.

Expected result

- The gate fails and the degraded index is not published; Q-01 is still answered from the previous index.
- The maintainer receives an alert with the evaluation report.

UAT-22 Disclaimer, coverage, model name and independence notice shown

Story: US-11. **Requirements:** NFR-07.1, NFR-07.2, NFR-07.3, NFR-07.4. **Priority:** High. **Status:** Not run.

Preconditions: None.

Steps

1. Open the assistant and ask Q-01.
2. Look for the disclaimer, coverage period, last update date, model name and version, and independence notice.

Expected result

- All five items are visible on the page or with the answer, in plain language.
- The disclaimer states that the service is not the official record, can make mistakes, and that the official minutes on eSCRIBE prevail.

UAT-23 Keyboard-only use

Story: US-21. **Requirements:** NFR-05.1. **Priority:** High. **Status:** Not run.

Preconditions: Desktop browser; mouse not used.

Steps

1. Use Tab, Shift+Tab, Enter and Space to reach every control.
2. Ask Q-01, open a citation, set a date range and use "Report a problem".

Expected result

- Every function can be completed with the keyboard in a logical order.
- The focus indicator is always visible, and there is no keyboard trap.

UAT-24 Screen reader labels and announcements

Story: US-21. **Requirements:** NFR-05.3, NFR-05.4. **Priority:** High. **Status:** Not run.

Preconditions: NVDA with Firefox or Chrome, and VoiceOver with Safari.

Steps

1. Navigate to the question box and the date fields.
2. Ask Q-01 and wait for the answer.
3. Move through the answer, citations and excerpts.

Expected result

- Every input has a spoken label.
- The progress status and the arrival of the answer are announced without moving focus unexpectedly.
- Citations and excerpts are read in a logical order with meaningful link text.

UAT-25 Contrast, zoom and reflow

Story: US-21. **Requirements:** NFR-05.2, NFR-05.5. **Priority:** High. **Status:** Not run.

Preconditions: Contrast checking tool available.

Steps

1. Check text and interface component contrast on the main page and an answer.
2. Zoom to 200%.
3. Narrow the window to 320 CSS pixels and ask Q-01.

Expected result

- Text contrast is at least 4.5 to 1, and large text and interface components at least 3 to 1.
- At 200% zoom and at 320 pixels wide, no content or function is lost and text does not need horizontal scrolling.

UAT-26 No account, no tracking, no identifiers in logs

Story: US-22. **Requirements:** NFR-06.1, NFR-06.2, NFR-06.3, NFR-06.4. **Priority:** High. **Status:** Not run.

Preconditions: Browser developer tools available; developer support to show log entries; privacy reviewer present.

Steps

1. Use the assistant and inspect cookies and network requests.
2. Read the privacy notice near the question box.
3. Ask Q-01, then ask the developer to show the matching log entry.
4. Ask the developer to show the retention job configuration and its last run.

Expected result

- No sign-in, no tracking cookies, no third-party analytics requests.
- The privacy notice tells users not to enter personal information and states what is stored and for how long.
- The log entry contains the question text, date and quality data only, with no IP address or other identifier.
- Log entries older than the retention period are deleted.

UAT-27 Response time and progress indicator

Story: US-23. **Requirements:** NFR-03.1, NFR-03.2, NFR-03.3. **Priority:** Medium. **Status:** Not run.

Preconditions: PA-05 and PA-06 results available for the snapshot. Tester acts as P4 City staff.

Steps

1. Ask five questions from the question bank and time each answer.
2. Load the search-only demo and time the results for one query after the page has loaded.
3. Watch for the progress indicator while an answer is prepared.

Expected result

- Observed times are consistent with PA-05 (p95 under 10 seconds) and PA-06 (under 1 second).
- A progress indicator is shown until each answer appears.

UAT-28 Spending cap switches to search-only mode

Story: US-24. **Requirements:** NFR-04.1. **Priority:** High. **Status:** Not run.

Preconditions: UAT instance with the daily cap set to a test value that has been reached.

Steps

1. Ask Q-01.

Expected result

- Search results with citations are returned, without a generated answer.
- A notice explains that generated answers are temporarily unavailable.

UAT-29 Staff copies a citation into a reply

Story: US-25. **Requirements:** FR-02.1, FR-02.2. **Priority:** Medium. **Status:** Not run.

Preconditions: Tester acts as P4 City staff with an email draft open.

Steps

1. Ask Q-01.
2. Choose "Copy citation" on the first citation.
3. Paste into the email draft.

Expected result

- The pasted text contains the meeting name, meeting date, page or item and the official link, as plain text.
- A confirmation is shown and announced to assistive technology.

UAT-30 Report a wrong answer without personal information

Story: US-26. **Requirements:** TR-07, NFR-06.1. **Priority:** High. **Status:** Not run.

Preconditions: Triage queue access for the maintainer; privacy reviewer present.

Steps

1. Ask Q-01.
2. Choose "Report a problem" and describe a test issue.
3. Check the maintainer's triage queue.

Expected result

- The report is submitted without asking for a name or contact details.
- The triage queue shows the report with the question, answer and citations.
- The report is acknowledged within the TR-07 target during the pilot.

14. Automated checks in Project A

These checks belong to the implementation's evaluation suite in Project A. The measured results below are copied exactly from Project A's README and `eval/results`, and were recorded in this pack on 18 October 2026. This pack does not re-measure them. Automated results do not replace user acceptance: all 30 UAT test cases remain "Not run", because no human UAT has been performed.

Summary: 13 of the 14 automated checks met their thresholds (PA-02 is report-only). PA-05, the p95 answer time for NFR-03, did not.

14.1 Results by check

Check ID	Automated check	Requirements	Threshold	Related UAT	Measured result (Project A)	Status
PA-01	Retrieval evaluation: hit@5 on answerable questions, overall and for exact-term and paraphrase subsets; keyword-only, semantic-only and combined	NFR-01, NFR-01.1, NFR-01.3, FR-06.1, FR-06.2, FR-06.3	At least 85%	UAT-13, UAT-14, UAT-15, UAT-21	Hybrid hit@5 0.954 (95% CI 0.87 to 0.98) on 65 answerable questions; BM25 0.923 (exact-term questions 52 of 52, paraphrased 8 of 13); dense 0.923	Met
PA-02	Retrieval evaluation by year and by format (PDF, HTML)	NFR-01.3	Report only	UAT-21	Hybrid hit@5 by year: 2023 0.966, 2024 0.917, 2025 1.000, 2026 0.917. By format: HTML only; PDF not applicable because of C-03 (Cloudflare challenge on PDF downloads, not bypassed)	Reported

Check ID	Automated check	Requirements	Threshold	Related UAT	Measured result (Project A)	Status
PA-03	Answer evaluation: share of citations that support their claim	NFR-02, NFR-02.1, NFR-02.2, FR-02.4, FR-01.2	At least 90%	UAT-04	90.2% of citations point to the gold agenda item; 80.5% to the exact gold passage; 92.7% support a statement on a read-through by an AI assistant (not a human review)	Met (by agenda item)
PA-04	Answer evaluation: share of unanswerable questions declined	NFR-02, NFR-02.2, FR-03.1	At least 80%	UAT-06, UAT-08	20 of 20 unanswerable questions declined (100%)	Met
PA-05	Latency benchmark: p95 end-to-end answer time	NFR-03, NFR-03.1	Under 10 seconds	UAT-27	p95 answer time 12.82 s with 6 passages; 10.96 s with 4 (local Qwen3.5-9B on an Apple M3 Pro)	Not met
PA-06	Search-only demo: time to results after page load	NFR-03, NFR-03.2	Under 1 second	UAT-27	28 to 35 ms per search after load; index ready 53 ms after navigation; query model 4.3 s on a first visit	Met
PA-07	Cost report: average model cost per answer from API usage, in CAD	NFR-04, NFR-04.2	Under CAD $0.05	UAT-28	USD 0 per answer (local model)	Met
PA-08	Automated accessibility scan of the UI (for example axe-core)	NFR-05, NFR-05.2, NFR-05.3	No serious or critical violations	UAT-23, UAT-24, UAT-25	axe-core: 0 violations (desktop light, desktop dark, 390 px)	Met (automated part)
PA-09	Privacy test: application logs contain no IP address or other identifier	NFR-06, NFR-06.2	Pass	UAT-26	Pass: no IP address or identifier in logs; server access log disabled	Met (automated part)
PA-10	UI test: disclaimer, coverage dates, model name and independence notice rendered	NFR-07, NFR-07.1, NFR-07.2, NFR-07.3, NFR-07.4	Pass	UAT-22	Pass	Met
PA-11	Citation tests: every answer other than "not found" has at least one citation with meeting name, date, page or item and a valid official URL	FR-02, FR-02.1, FR-02.2, FR-02.3	100% of answers	UAT-02, UAT-03, UAT-12	82 of 82 citations valid, each with meeting name, date, agenda item and an official URL	Met (agenda-item citations)
PA-12	Date filter tests: no retrieved passage outside the selected range	FR-04, FR-04.3	100% of filtered queries	UAT-16	0 passages outside the selected date range	Met
PA-13	Excerpt tests: every published excerpt is at most 80 characters and a verbatim substring of its source	FR-05, FR-05.1	100% of excerpts	UAT-11	All 1,371 published excerpts are verbatim and at most 80 characters (decisions only); published excerpt share 7.8% of the parsed text	Met
PA-14	Ingestion tests: re-run creates no duplicates; log written; coverage updated	FR-07, FR-07.2, FR-07.3	Pass	UAT-19, UAT-20	Pass	Met

14.2 Results by requirement

Requirement	Status	Based on	Notes and next steps
FR-01	Met	PA-03	Answers cite their sources; 90.2% of citations point to the gold agenda item. Human UAT not run.

Requirement	Status	Based on	Notes and next steps
FR-02	Met with agenda-item citations	PA-03, PA-11	Citations give the agenda item instead of a page number, because PDF downloads are blocked (C-03). Page citations depend on an agreed access method (OQ-05).
FR-03	Met	PA-04	20 of 20 unanswerable questions declined.
FR-04	Met	PA-12	No passage outside the selected date range.
FR-05	Met	PA-13	Published excerpts are at most 80 characters, within the 300-character limit.
FR-06	Met	PA-01	Hybrid hit@5 0.954. Subset results are reported for BM25 only (exact-term 52 of 52, paraphrased 8 of 13), so the combined retriever's paraphrase subset (FR-06.2) is still to be reported.
FR-07	Met	PA-14	Re-run creates no duplicates; log written; coverage updated.
NFR-01	Met	PA-01, PA-02	Hybrid hit@5 0.954 against the 85% target, on HTML minutes only.
NFR-02	Met	PA-03, PA-04	Citation target met by agenda item (90.2%) and refusal target met (100%). The 92.7% support figure comes from an AI read-through, not a human review, so UAT-04 remains the human check (NFR-02.1).
NFR-03	Partly met	PA-05, PA-06	Search-only time met. The p95 answer time was not met: 12.82 s with 6 passages and 10.96 s with 4. Next steps: retrieve fewer passages, use a smaller model or use a hosted model, then re-measure PA-05.
NFR-04	Met	PA-07	USD 0 per answer with a local model. Re-measure cost if a hosted model is adopted to meet NFR-03.
NFR-05	Automated part met	PA-08	Manual keyboard and screen reader checks remain (UAT-23 to UAT-25).
NFR-06	Automated part met	PA-09	Retention (NFR-06.3) is checked in UAT-26; the processing confirmation (NFR-06.5) is a go-live condition.
NFR-07	Met	PA-10	Disclaimer, coverage dates, model name and independence notice rendered.

Automated scans find only part of the accessibility issues, so the manual cases UAT-23 to UAT-25 remain required.

15. Reporting and sign-off

Daily during execution: test cases run, passed, failed and blocked; new defects by severity; blockers.

UAT summary report: scope tested; results by test case; defects by severity and status; Project A check results with versions; deviations from this plan; open risks; go/no-go recommendation.

Role	Name	Decision (accept, accept with conditions, reject)	Signature	Date
Business sponsor				
Product owner				
UAT lead				
Privacy reviewer (ATIPP Coordinator)				

Requirements Traceability Matrix: St. John's Council Minutes Assistant

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Item	Value
Document ID	CMA-BA-04
Version	0.9.1 (generated; Project A results recorded 18 October 2026)
Author	Mostafa Rafiur Wasib
Date	17 October 2026
Machine-readable version	traceability.csv (one row per requirement and per detailed requirement)
How it is produced	Generated by `tools/trace/build-traceability.mjs` from the BRD, the user stories and the UAT plan. The build fails if any reference is broken or any requirement is uncovered. Do not edit this file by hand.

1. How to read this matrix

Each row follows one chain: **business requirement** (why) to **functional or non-functional requirement** (what and how well) to **user stories** (how it is delivered) to **UAT test cases** (how the business accepts it) to **Project A automated checks** (how it is measured on every build). Requirement IDs are defined in 01-business-requirements-document.md, stories in 02-user-stories.md, and UAT cases and automated checks (PA-nn) in 03-uat-test-plan.md.

Status is the measured outcome from Project A for each requirement, recorded on 18 October 2026 from Project A's README and `eval/results` (UAT plan section 14). The figures are Project A's; this pack does not re-measure them. Automated results do not replace user acceptance: every UAT test case is still "Not run".

2. Coverage summary

Item	Count
Business requirements	3
Stakeholder requirements	11
Functional requirements (detailed)	7 (27)
Non-functional requirements (detailed)	7 (25)
Transition requirements	11
Business rules	11
User stories (points)	26 (90); Release 1: 19 stories, 69 points; Release 2: 7 stories, 21 points
UAT test cases	30 (22 High, 8 Medium), all Not run
Project A automated checks	14: 13 met their thresholds (PA-02 report-only), 1 not met (PA-05)

Checks performed by the build (all passed):

- Every one of the 14 functional and non-functional requirements traces to at least one business requirement, one user story and one UAT test case.

- Every one of the 26 user stories is verified by at least one UAT test case, and its "Verified by" list matches the UAT plan.
- Every UAT test case and automated check references valid stories and requirements, and each requirement a UAT case tests is satisfied by one of its stories.
- Summary tables and detailed sections agree on priority, estimate, release and references.
- 50 of 52 detailed requirements are verified directly by a UAT case or an automated check. The other 2 are listed in section 7 with how they are verified.
- Every check has a measured result and a valid status, and each requirement's status agrees with its checks (a requirement with a check that is not met cannot be marked met).

3. Forward traceability

Business requirement	Requirement	Priority	User stories	UAT test cases	Project A checks	Threshold	Status
BR-01, BR-03	**FR-01** Answer natural-language questions about Regular Council Meeting minutes	Must	US-01, US-04, US-05, US-06	UAT-01, UAT-04, UAT-05, UAT-09, UAT-10	PA-03	PA-03: At least 90%	Met
BR-02	**FR-02** Every answer cites meeting name, meeting date and page, and links to the official document	Must	US-07, US-08, US-09, US-12, US-25	UAT-02, UAT-03, UAT-04, UAT-12, UAT-13, UAT-29	PA-03, PA-11	PA-03: At least 90%; PA-11: 100% of answers	Met with agenda-item citations
BR-02	**FR-03** When the minutes do not contain the answer, say so instead of guessing	Must	US-02, US-03	UAT-06, UAT-07, UAT-08	PA-04	PA-04: At least 80%	Met
BR-01	**FR-04** Filter by meeting date range	Should	US-14	UAT-16, UAT-17	PA-12	PA-12: 100% of filtered queries	Met
BR-02	**FR-05** Show the retrieved source passages (short excerpts) next to the answer	Must	US-10	UAT-11	PA-13	PA-13: 100% of excerpts	Met
BR-01	**FR-06** Hybrid retrieval (keyword + semantic) so exact terms (by-law numbers, street names) and paraphrases both work	Must	US-12, US-13	UAT-13, UAT-14, UAT-15	PA-01	PA-01: At least 85%	Met
BR-01, BR-02	**FR-07** Ingest newly published minutes without manual editing (repeatable pipeline)	Must	US-15, US-16, US-17, US-18	UAT-18, UAT-19, UAT-20	PA-14	PA-14: Pass	Met
BR-01, BR-02	**NFR-01** Quality gate (retrieval): correct source page in the top 5 results (hit@5) for at least 85% of answerable evaluation questions	Must	US-19	UAT-21	PA-01, PA-02	PA-01: At least 85%; PA-02: Report only	Met
BR-02	**NFR-02** Quality gate (answers): at least 90% of answer citations point to a passage that supports the claim; at least 80% of unanswerable questions correctly declined	Must	US-20	UAT-04	PA-03, PA-04	PA-03: At least 90%; PA-04: At least 80%	Met

Business requirement	Requirement	Priority	User stories	UAT test cases	Project A checks	Threshold	Status
BR-01	**NFR-03** Performance: p95 end-to-end answer time under 10 seconds; search-only (demo) under 1 second after the page loads	Should	US-23	UAT-27	PA-05, PA-06	PA-05: Under 10 seconds; PA-06: Under 1 second	Partly met
BR-03	**NFR-04** Cost: average model cost per answer under CAD $0.05 (measured from API usage)	Must	US-24	UAT-28	PA-07	PA-07: Under CAD $0.05	Met
BR-01, BR-02	**NFR-05** Accessibility: demo UI meets WCAG 2.1 AA (keyboard use, contrast, labels)	Must	US-21	UAT-23, UAT-24, UAT-25	PA-08	PA-08: No serious or critical violations	Automated part met
BR-02	**NFR-06** Privacy: no accounts, no personal data collected; questions are not stored with identifiers (aligned with NL ATIPPA, 2015 principles)	Must	US-22, US-26	UAT-26, UAT-30	PA-09	PA-09: Pass	Automated part met
BR-02	**NFR-07** Transparency: disclaimer, data coverage dates and model name visible in the UI	Must	US-11, US-17	UAT-20, UAT-22	PA-10	PA-10: Pass	Met

4. Measured results from Project A

Recorded on 18 October 2026 from Project A's README and `eval/results`. 13 of the 14 automated checks met their thresholds (PA-02 is report-only); PA-05 did not. Full detail is in UAT plan section 14.

By check

Check	Requirements	Threshold	Measured result (Project A)	Status
PA-01	FR-06, NFR-01	At least 85%	Hybrid hit@5 0.954 (95% CI 0.87 to 0.98) on 65 answerable questions; BM25 0.923 (exact-term questions 52 of 52, paraphrased 8 of 13); dense 0.923	Met
PA-02	NFR-01	Report only	Hybrid hit@5 by year: 2023 0.966, 2024 0.917, 2025 1.000, 2026 0.917. By format: HTML only; PDF not applicable because of C-03 (Cloudflare challenge on PDF downloads, not bypassed)	Reported
PA-03	FR-01, FR-02, NFR-02	At least 90%	90.2% of citations point to the gold agenda item; 80.5% to the exact gold passage; 92.7% support a statement on a read-through by an AI assistant (not a human review)	Met (by agenda item)
PA-04	FR-03, NFR-02	At least 80%	20 of 20 unanswerable questions declined (100%)	Met
PA-05	NFR-03	Under 10 seconds	p95 answer time 12.82 s with 6 passages; 10.96 s with 4 (local Qwen3.5-9B on an Apple M3 Pro)	Not met
PA-06	NFR-03	Under 1 second	28 to 35 ms per search after load; index ready 53 ms after navigation; query model 4.3 s on a first visit	Met
PA-07	NFR-04	Under CAD $0.05	USD 0 per answer (local model)	Met
PA-08	NFR-05	No serious or critical violations	axe-core: 0 violations (desktop light, desktop dark, 390 px)	Met (automated part)
PA-09	NFR-06	Pass	Pass: no IP address or identifier in logs; server access log disabled	Met (automated part)

Check	Requirements	Threshold	Measured result (Project A)	Status
PA-10	NFR-07	Pass	Pass	Met
PA-11	FR-02	100% of answers	82 of 82 citations valid, each with meeting name, date, agenda item and an official URL	Met (agenda-item citations)
PA-12	FR-04	100% of filtered queries	0 passages outside the selected date range	Met
PA-13	FR-05	100% of excerpts	All 1,371 published excerpts are verbatim and at most 80 characters (decisions only); published excerpt share 7.8% of the parsed text	Met
PA-14	FR-07	Pass	Pass	Met

By requirement

Requirement	Status	Based on	Notes and next steps
FR-01	Met	PA-03	Answers cite their sources; 90.2% of citations point to the gold agenda item. Human UAT not run.
FR-02	Met with agenda-item citations	PA-03, PA-11	Citations give the agenda item instead of a page number, because PDF downloads are blocked (C-03). Page citations depend on an agreed access method (OQ-05).
FR-03	Met	PA-04	20 of 20 unanswerable questions declined.
FR-04	Met	PA-12	No passage outside the selected date range.
FR-05	Met	PA-13	Published excerpts are at most 80 characters, within the 300-character limit.
FR-06	Met	PA-01	Hybrid hit@5 0.954. Subset results are reported for BM25 only (exact-term 52 of 52, paraphrased 8 of 13), so the combined retriever's paraphrase subset (FR-06.2) is still to be reported.
FR-07	Met	PA-14	Re-run creates no duplicates; log written; coverage updated.
NFR-01	Met	PA-01, PA-02	Hybrid hit@5 0.954 against the 85% target, on HTML minutes only.
NFR-02	Met	PA-03, PA-04	Citation target met by agenda item (90.2%) and refusal target met (100%). The 92.7% support figure comes from an AI read-through, not a human review, so UAT-04 remains the human check (NFR-02.1).
NFR-03	Partly met	PA-05, PA-06	Search-only time met. The p95 answer time was not met: 12.82 s with 6 passages and 10.96 s with 4. Next steps: retrieve fewer passages, use a smaller model or use a hosted model, then re-measure PA-05.
NFR-04	Met	PA-07	USD 0 per answer with a local model. Re-measure cost if a hosted model is adopted to meet NFR-03.
NFR-05	Automated part met	PA-08	Manual keyboard and screen reader checks remain (UAT-23 to UAT-25).
NFR-06	Automated part met	PA-09	Retention (NFR-06.3) is checked in UAT-26; the processing confirmation (NFR-06.5) is a go-live condition.
NFR-07	Met	PA-10	Disclaimer, coverage dates, model name and independence notice rendered.

5. Business requirement view

Business requirement	Functional requirements	Non-functional requirements	User stories	UAT test cases
BR-01 Reduce the time it takes residents and staff to find what council decided on a topic.	FR-01, FR-04, FR-06, FR-07	NFR-01, NFR-03, NFR-05	US-01, US-04, US-05, US-06, US-12, US-13, US-14, US-15, US-16, US-17, US-18, US-19, US-21, US-23	UAT-01, UAT-04, UAT-05, UAT-09, UAT-10, UAT-13, UAT-14, UAT-15, UAT-16, UAT-17, UAT-18, UAT-19, UAT-20, UAT-21, UAT-23, UAT-24, UAT-25, UAT-27

Business requirement	Functional requirements	Non-functional requirements	User stories	UAT test cases
BR-02 Increase trust and transparency: every answer is traceable to the official record.	FR-02, FR-03, FR-05, FR-07	NFR-01, NFR-02, NFR-05, NFR-06, NFR-07	US-02, US-03, US-07, US-08, US-09, US-10, US-11, US-12, US-15, US-16, US-17, US-18, US-19, US-20, US-21, US-22, US-25, US-26	UAT-02, UAT-03, UAT-04, UAT-06, UAT-07, UAT-08, UAT-11, UAT-12, UAT-13, UAT-18, UAT-19, UAT-20, UAT-21, UAT-22, UAT-23, UAT-24, UAT-25, UAT-26, UAT-29, UAT-30
BR-03 Reduce routine "what did council decide about X" enquiries handled manually (assumption to validate).	FR-01	NFR-04	US-01, US-04, US-05, US-06, US-24	UAT-01, UAT-04, UAT-05, UAT-09, UAT-10, UAT-28

6. Backward traceability

Read from right to left: each UAT case, the stories it accepts, and the requirements and business requirements behind them.

UAT test case	Priority	User stories	Requirements	Business requirements
UAT-01 Answer a plain-language question with citations	High	US-01	FR-01.1, FR-01.2	BR-01, BR-03
UAT-02 Citation shows meeting name, date and page	High	US-07	FR-02.1	BR-02
UAT-03 Citation link opens the official document at the cited page	High	US-08	FR-02.2	BR-02
UAT-04 Every statement is supported by its cited passage	High	US-01, US-07, US-10, US-20	FR-01.2, FR-02.4, NFR-02.1	BR-01, BR-02, BR-03
UAT-05 Outcome reported exactly as recorded	High	US-05	FR-01.4	BR-01, BR-03
UAT-06 "Not found" response for an unanswerable question	High	US-02	FR-03.1	BR-02
UAT-07 Minutes not yet published	Medium	US-02	FR-03.3	BR-02
UAT-08 Out-of-scope questions declined	High	US-03	FR-03.2	BR-02
UAT-09 Committee of the Whole question explained	Medium	US-06	FR-01.5	BR-01, BR-03
UAT-10 Several decisions listed in date order	Medium	US-04	FR-01.3	BR-01, BR-03
UAT-11 Source excerpts shown with the answer	High	US-10	FR-05.1, FR-05.2, FR-05.3	BR-02
UAT-12 HTML-only minutes cited by agenda item	Medium	US-09	FR-02.3	BR-02
UAT-13 Resolution number retrieves the right passage	High	US-12	FR-06.1, FR-02.5	BR-01, BR-02
UAT-14 File number and civic address retrieve the right passage	High	US-12	FR-06.1	BR-01
UAT-15 Paraphrased question retrieves the right passage	High	US-13	FR-06.2	BR-01
UAT-16 Date range restricts the sources	Medium	US-14	FR-04.1, FR-04.3	BR-01
UAT-17 Invalid date range rejected with a clear message	Medium	US-14	FR-04.2	BR-01
UAT-18 New minutes ingested on schedule without manual editing	High	US-15	FR-07.1, FR-07.4	BR-01, BR-02

UAT test case	Priority	User stories	Requirements	Business requirements
UAT-19 Re-run creates no duplicates; changed minutes re-indexed	High	US-16, US-18	FR-07.2, FR-07.5	BR-01, BR-02
UAT-20 Ingestion log and coverage dates updated	High	US-17	FR-07.3, NFR-07.2	BR-01, BR-02
UAT-21 Failed quality gate keeps the previous index	High	US-19	NFR-01.2	BR-01, BR-02
UAT-22 Disclaimer, coverage, model name and independence notice shown	High	US-11	NFR-07.1, NFR-07.2, NFR-07.3, NFR-07.4	BR-02
UAT-23 Keyboard-only use	High	US-21	NFR-05.1	BR-01, BR-02
UAT-24 Screen reader labels and announcements	High	US-21	NFR-05.3, NFR-05.4	BR-01, BR-02
UAT-25 Contrast, zoom and reflow	High	US-21	NFR-05.2, NFR-05.5	BR-01, BR-02
UAT-26 No account, no tracking, no identifiers in logs	High	US-22	NFR-06.1, NFR-06.2, NFR-06.3, NFR-06.4	BR-02
UAT-27 Response time and progress indicator	Medium	US-23	NFR-03.1, NFR-03.2, NFR-03.3	BR-01
UAT-28 Spending cap switches to search-only mode	High	US-24	NFR-04.1	BR-03
UAT-29 Staff copies a citation into a reply	Medium	US-25	FR-02.1, FR-02.2	BR-02
UAT-30 Report a wrong answer without personal information	High	US-26	TR-07, NFR-06.1	BR-02

7. Detailed requirements without direct verification

These detailed requirements are not referenced directly by a UAT case or an automated check. Their parent requirement is covered, and the note says how each one is verified.

Detailed requirement	Priority	Requirement	How it is verified
NFR-05.6	Could	Answers use plain language (target: grade 8 reading level or lower, excluding quoted names and titles).	Could priority. Verified by a readability check of the question bank answers during UAT-04 review; not an exit criterion.
NFR-06.5	Must	Document where question text is processed (provider, region, retention and training terms) and confirm it with the ATIPP Coordinator before any production use.	Verified as a go-live condition: the ATIPP Coordinator confirms the processing arrangements before production use (TR-09).

8. Stakeholder requirements

Stakeholder requirement	Stakeholder	Business requirements	Satisfied by	User stories
SR-01 Ask about a council decision in everyday words, without knowing the meeting date or the wording used in the minutes.	Residents	BR-01	FR-01, FR-06	US-01, US-04, US-05, US-06, US-12, US-13
SR-02 Check any answer against the official minutes quickly: meeting, date, page and a working link.	All users	BR-02	FR-02, FR-05	US-07, US-08, US-09, US-10, US-12, US-25
SR-03 Quote the record accurately: see the exact passage, the recorded outcome and where it appears.	Journalists and researchers	BR-02	FR-01, FR-02, FR-05	US-01, US-04, US-05, US-06, US-07, US-08, US-09, US-10, US-12, US-25

Stakeholder requirement	Stakeholder	Business requirements	Satisfied by	User stories
SR-04 Be told plainly when the minutes do not answer the question, and what to do next.	All users	BR-02	FR-03	US-02, US-03
SR-05 Find every decision about a place, application or plan within a period, in date order.	Community association volunteers	BR-01	FR-01, FR-04, FR-06	US-01, US-04, US-05, US-06, US-12, US-13, US-14
SR-06 Answer a routine decision enquiry quickly with a citable reference to the official minutes.	City staff answering enquiries	BR-03	FR-01, FR-02, NFR-03	US-01, US-04, US-05, US-06, US-07, US-08, US-09, US-12, US-23, US-25
SR-07 The service never presents itself as the official record and always points to eSCRIBE.	City Clerk's office	BR-02	FR-02, NFR-07	US-07, US-08, US-09, US-11, US-12, US-17, US-25
SR-08 No personal information is collected beyond what is necessary, and questions cannot be linked to people.	ATIPP Coordinator	BR-02	NFR-06	US-22, US-26
SR-09 Use the service with a keyboard, screen reader or magnification.	People with disabilities	BR-01	NFR-05	US-21
SR-10 Know how current the information is and how the answers are produced.	All users	BR-02	NFR-07, FR-07	US-11, US-15, US-16, US-17, US-18
SR-11 Keep content current without manual editing, and know when quality drops or costs rise.	Product owner and maintainer	BR-01, BR-02, BR-03	FR-07, NFR-01, NFR-02, NFR-04	US-15, US-16, US-17, US-18, US-19, US-20, US-24

9. Transition requirements

Transition requirements are verified through transition activities rather than product UAT, except where a story implements them.

Transition requirement	Priority	User stories	UAT test cases	Verified by
TR-01 Initial load: ingest all Regular Council Meeting minutes listed on eSCRIBE up to the cut-over date, and reconcile counts with the portal listing (method in Appendix A).	Must	None	None	Meetings with minutes listed equal meetings ingested, or each difference is explained in a reconciliation report.
TR-02 Launch baseline: run the full evaluation suite on the initial index and publish the results for NFR-01 to NFR-04.	Must	None	None	Report published with the evaluation set version, model and date.
TR-03 Pilot: four-week soft launch with representatives of each persona, a short feedback survey, and the SM-01 and SM-03 measurements.	Should	None	None	Pilot report with findings and go-forward decisions.
TR-04 Staff enablement: a 30-minute walkthrough and a one-page quick reference for staff who answer enquiries (verifying citations, limits, what to do after "not found").	Should	None	None	Material reviewed by the City Clerk's office; sessions held for all pilot staff.
TR-05 Public help content: "How to use", "What answers are based on", "Limitations" and "Privacy" pages in plain language.	Must	None	None	Pages published and checked for plain language and accessibility.
TR-06 Communications: launch message and FAQ with consistent key messages: a way to find minutes, not the official record; check the link; how to report a problem.	Should	None	None	Approved by Communications and the City Clerk's office.
TR-07 Support: a "report a problem" channel that needs no personal information, a named triage owner, and targets (proposed: acknowledge within 2 business days; correct or suppress a wrong answer within 5 business days).	Must	US-26	UAT-30	Process documented; a test report handled end to end (UAT-30).

Transition requirement	Priority	User stories	UAT test cases	Verified by
TR-08 Retire workarounds: list the informal aids used today (assumed: bookmarks, personal lists of frequently requested decisions, saved searches) and retire those the assistant replaces after the pilot. eSCRIBE remains the system of record and is not retired.	Could	None	None	Inventory and decision log agreed with the City Clerk's office.
TR-09 Go/no-go and rollback: go-live requires the UAT exit criteria (UAT plan section 5) and the privacy confirmation (NFR-06.5). Rollback restores the previous index or switches to search-only mode within one hour.	Must	None	None	Rollback rehearsed once before go-live.
TR-10 Benefits review: repeat the SM-01 to SM-05 measurements at the end of the pilot and three and six months after launch.	Should	None	None	Benefits report to the sponsor.
TR-11 Handover: a runbook for ingestion, evaluation and incident response, kept with the code.	Should	None	None	A second person can run the pipeline using only the runbook.

10. Updating results from Project A

The first results were recorded on 18 October 2026. After each new Project A evaluation run:

1. Copy each measured value and its status (Met, Not met or Reported) into UAT plan section 14.1, and update the requirement status and notes in section 14.2.
2. Re-run `npm run trace` in the `tools` folder. It regenerates `traceability.csv` and this matrix, and fails if a requirement's status contradicts its checks. Do not edit either file by hand.
3. Re-run `npm run summary` and `npm run pdf` so that the two-page summary and the full pack carry the same figures.

Appendix: Diagrams

The four diagrams in this pack, each on its own landscape page so that labels stay readable. In the documents they appear at reduced size with a link to these pages.

How the BPMN models were checked

- Generated as BPMN 2.0 XML with diagram interchange (DI) coordinates, so they open in bpmn.io and Camunda Modeler.
- Imported with bpmn-js in headless Chrome with zero import warnings, plus automated checks that no label wraps unexpectedly and no task-type icon overlaps its text.
- Linted with bpmnlint (recommended rules) with no findings, and validated against the official OMG BPMN 2.0 XML schema.
- The SVG images are exported by bpmn-js itself, so they match the models exactly.

Reading BPMN

Symbol	Meaning
Pool (large horizontal box)	A participant, such as the City or the assistant. A pool with no content is a "black box", here the eSCRIBE portal.
Lane (band inside a pool)	A role or system component inside a participant.
Thin circle / thick circle	Start event / end event. An envelope or clock inside marks a message or timer trigger.
Rounded rectangle	A task. A person icon marks a user task, gears a service (system) task, an envelope a send task.
Diamond with X	Exclusive gateway: exactly one outgoing path is taken. Unlabelled diamonds merge paths.
Solid arrow / dashed arrow	Sequence flow within a pool / message flow between pools.
Cylinder and dotted arrows	Data store (the minutes index) and the tasks that read or write it.
Square bracket with text	Annotation, used for pain points (PP-nn) and requirement references.

Diagram 1. Current-state process map (as-is), BPMN 2.0

Pools: information seeker; eSCRIBE public meetings portal (black box); City of St. John's with lanes for the City Clerk's office and 311, and the ATIPP Coordinator. City lanes are assumptions to validate. Annotations PP-01 to PP-05 mark pain points (BRD section 7.4). Source: diagrams/current-state.bpmn.

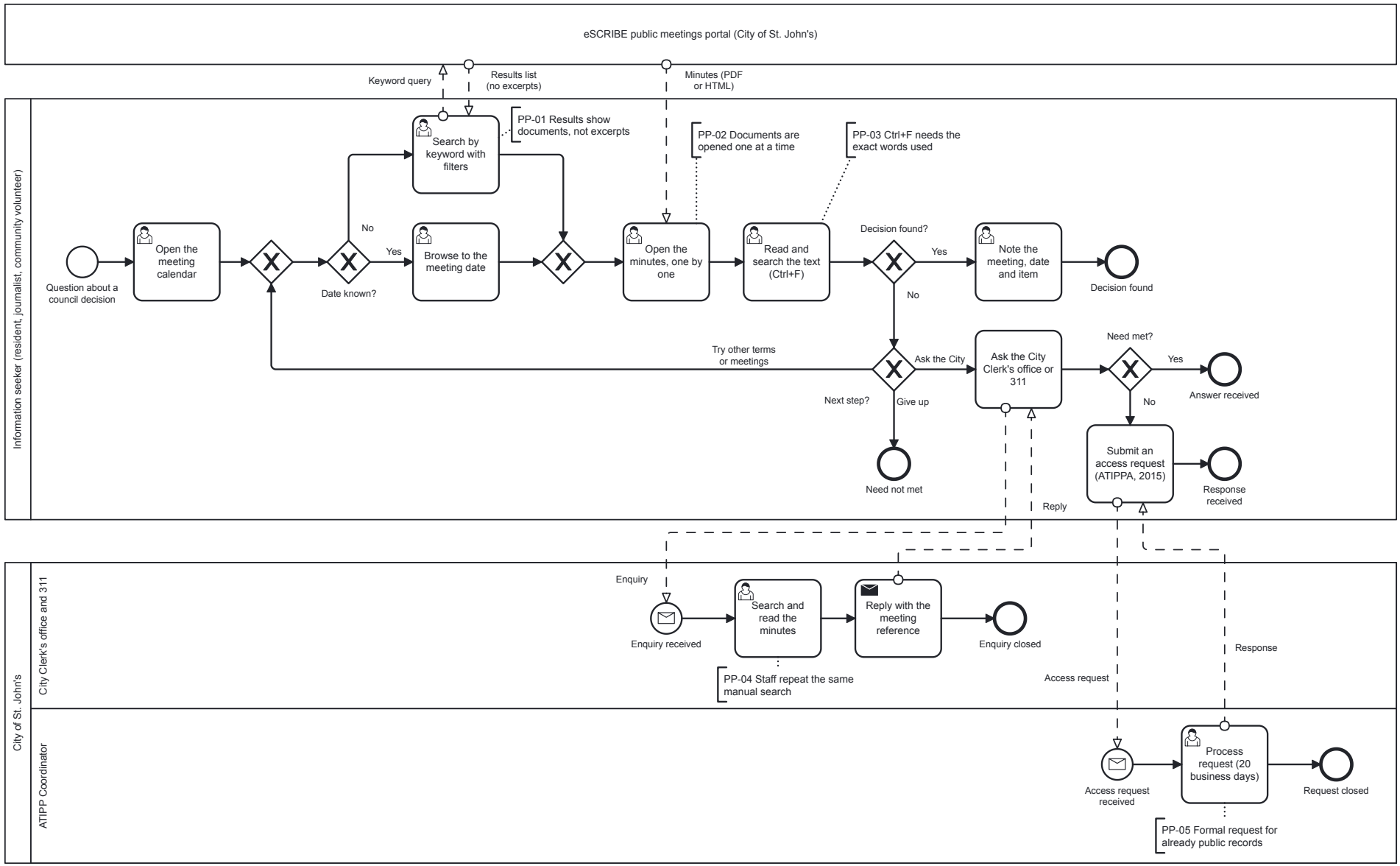


Diagram 2. Future-state process map (to-be), BPMN 2.0

Pools: City staff; resident, journalist or community volunteer; the assistant (answer service and ingestion pipeline lanes, sharing the minutes index); eSCRIBE portal (black box, unchanged). Annotations reference requirement IDs. Source: diagrams/future-state.bpmn.

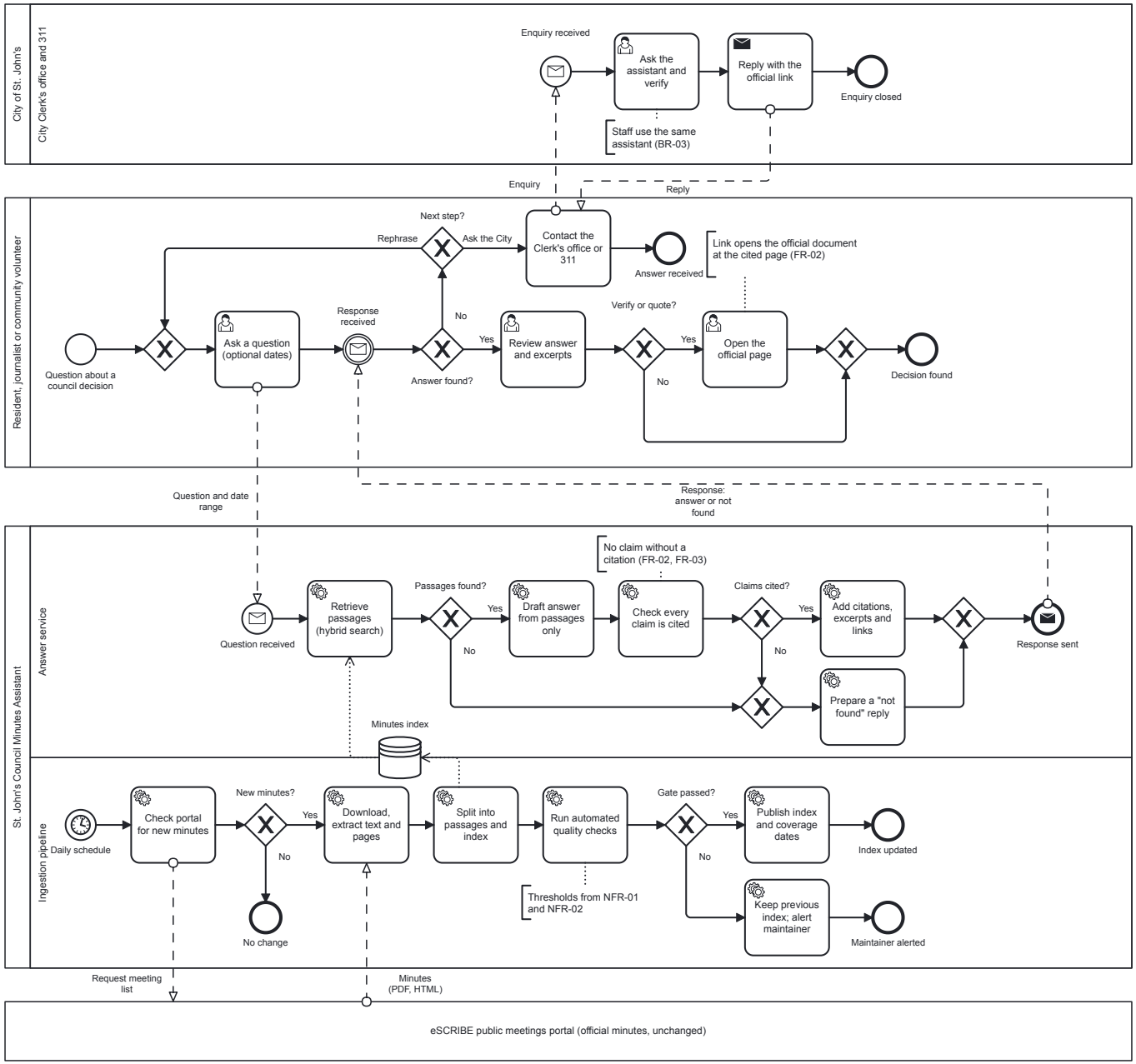
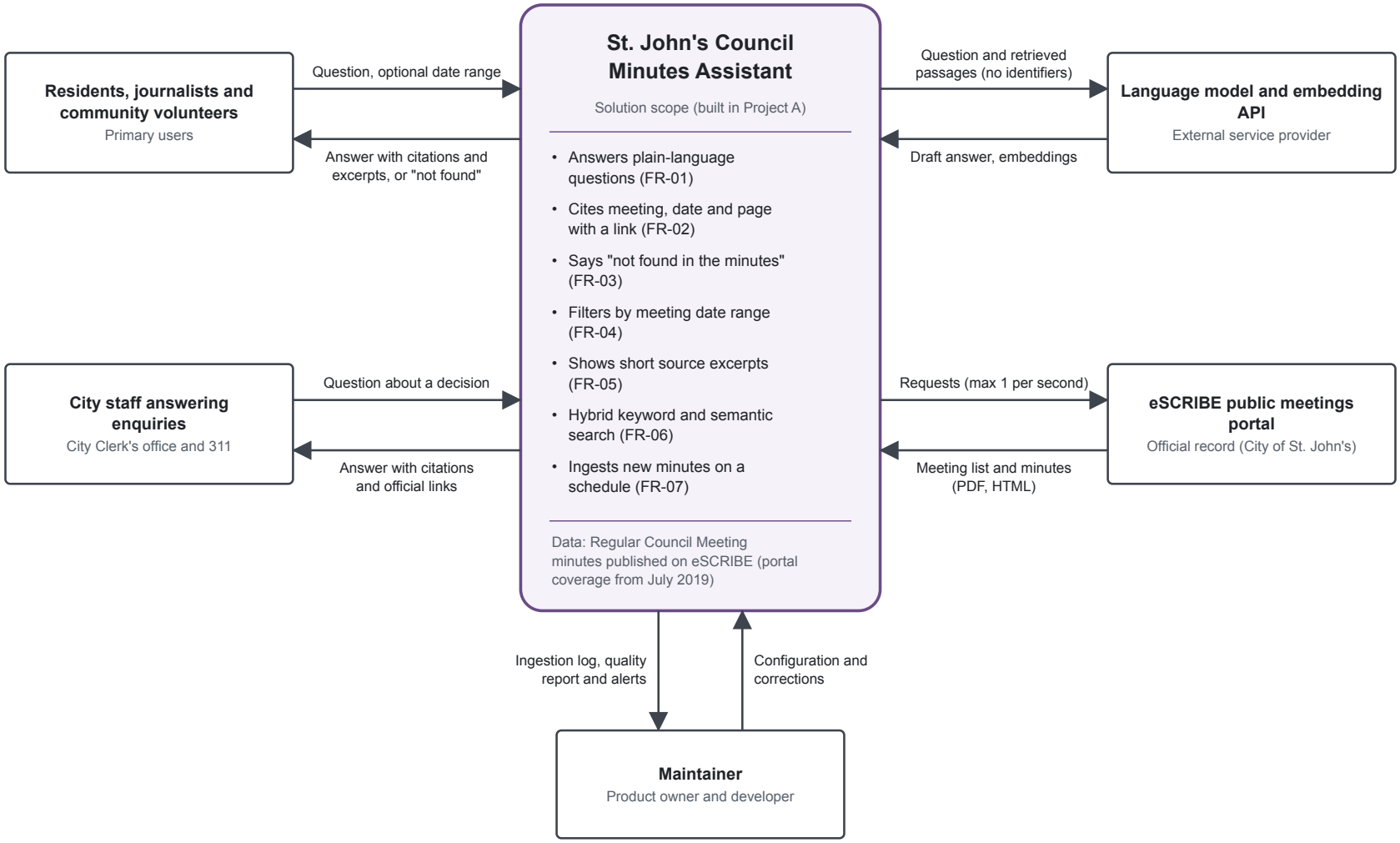


Diagram 3. Context diagram

External entities and the information exchanged with the solution (BRD section 4.4). Source: diagrams/context-diagram.svg.

Context diagram: St. John's Council Minutes Assistant

External entities and the information that flows between them and the solution. Arrows point to the receiver.

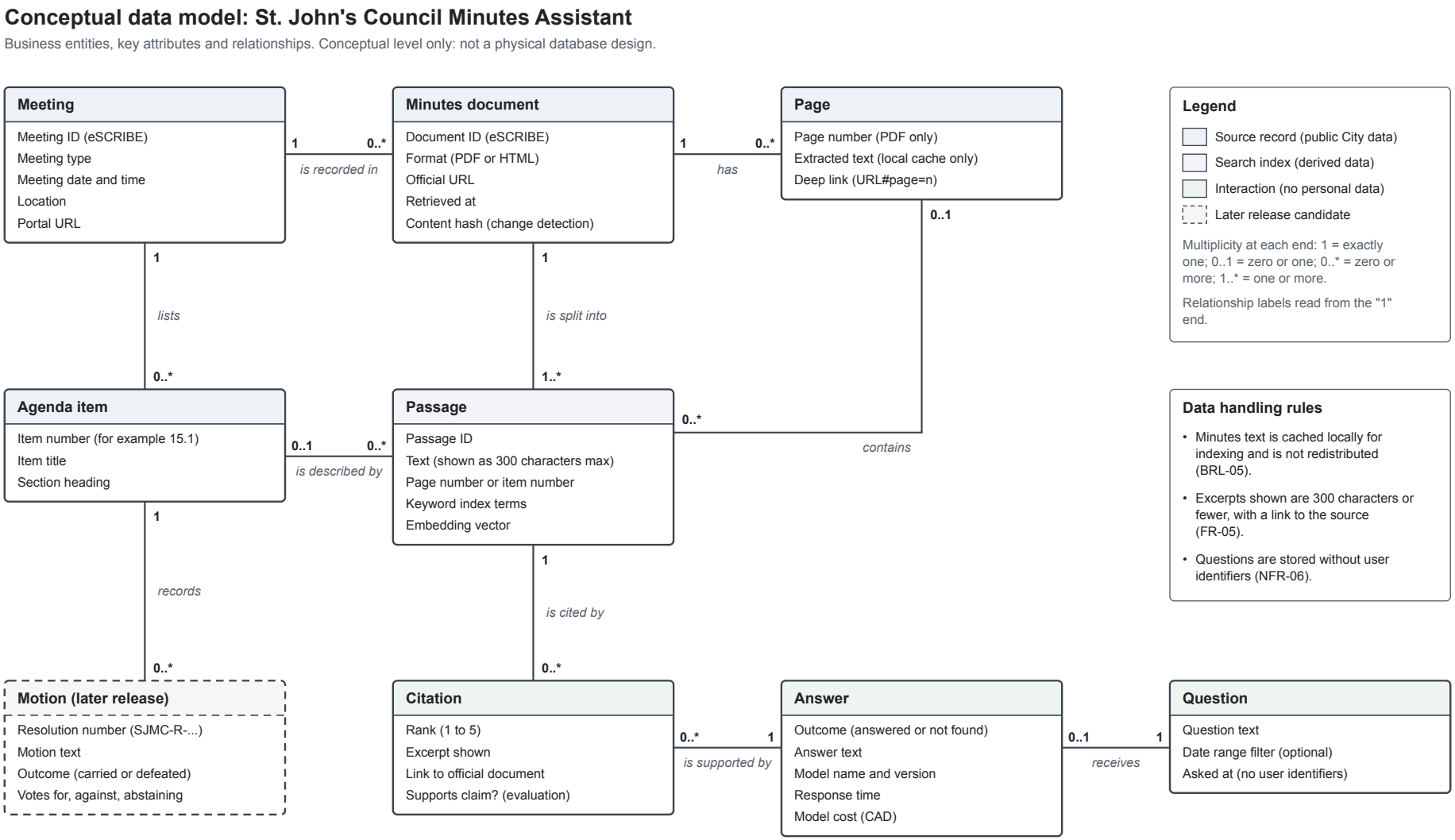


Out of scope: Committee of the Whole and other committee minutes, agendas, attachments, video, records not published on eSCRIBE, and any change to eSCRIBE itself. The eSCRIBE portal remains the official record.

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Diagram 4. Conceptual data model

Entities, key attributes and multiplicities (BRD section 10). Source: diagrams/conceptual-data-model.svg.



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